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(54) **SMALL VOLUME PROCESSING SYSTEMS
AND METHODS WITH CAPACITIVE
SENSING**

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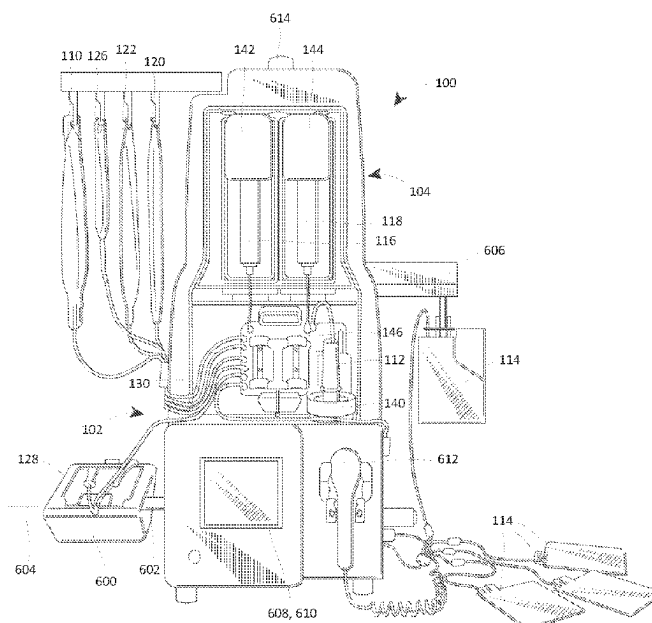
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(57) **ABSTRACT**

A fluid processing system may include a flow control cassette comprising at least one interface sensor chamber in fluid communication with at least one of a plurality of separate channels, the at least one interface sensor chamber defined at least in part by a wall, and at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber. The fluid processing system may include, in the alternative or in addition, at least one syringe comprising a wall defining a barrel having a first end and a second end, the barrel having a bore with or without a piston or plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the syringe.

8 Claims, 14 Drawing Sheets



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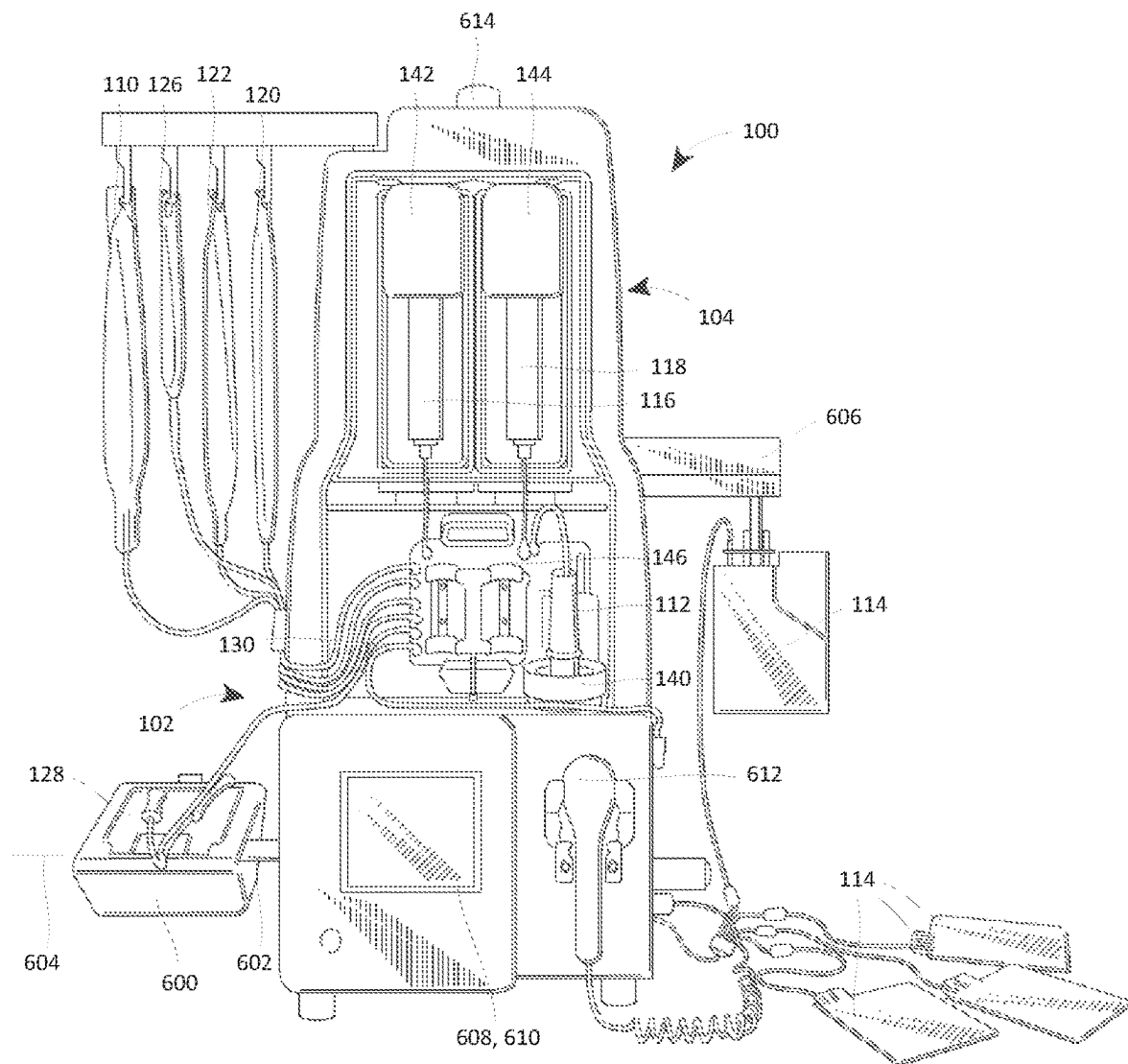


FIG. 1

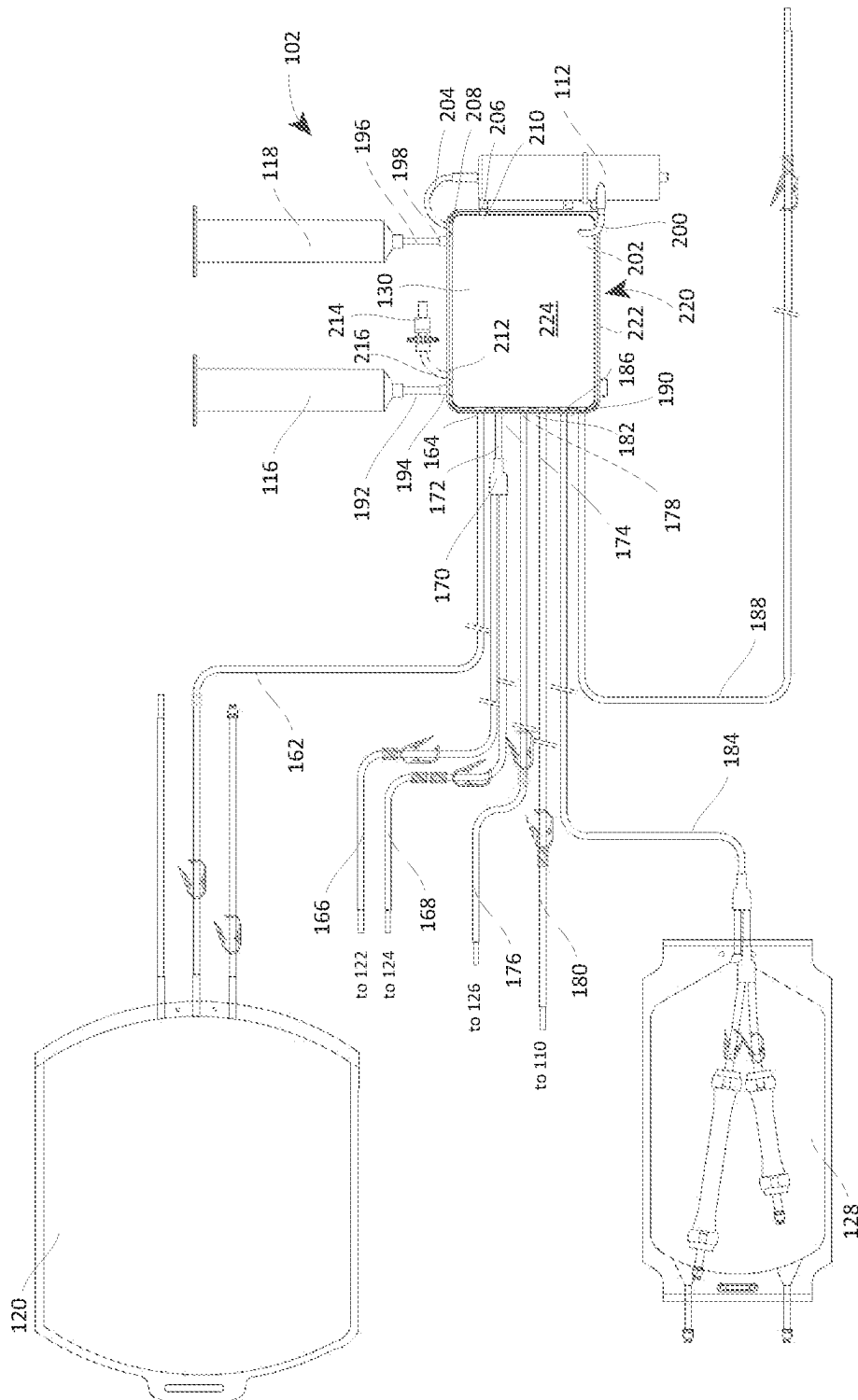


FIG. 2

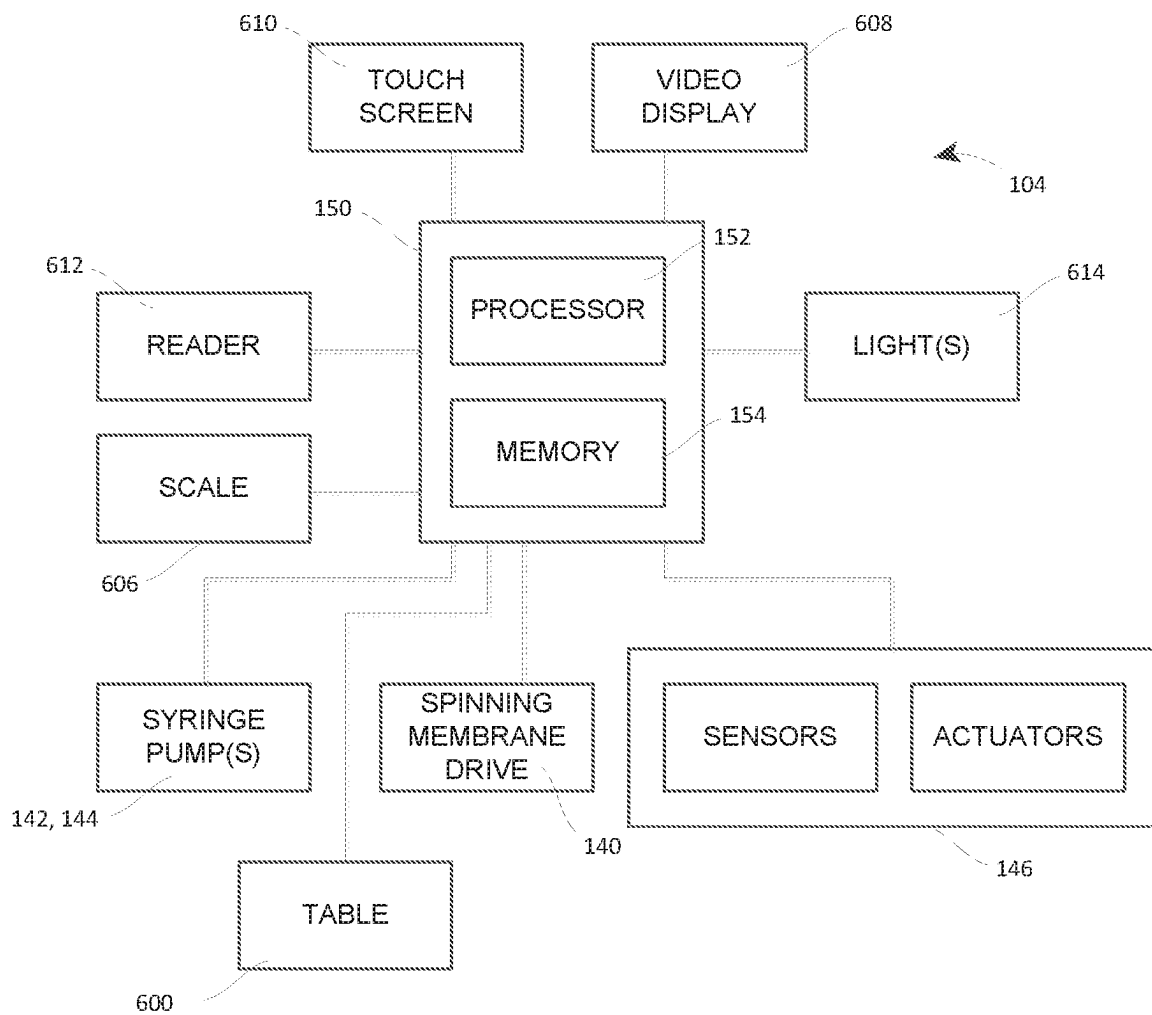


FIG. 3

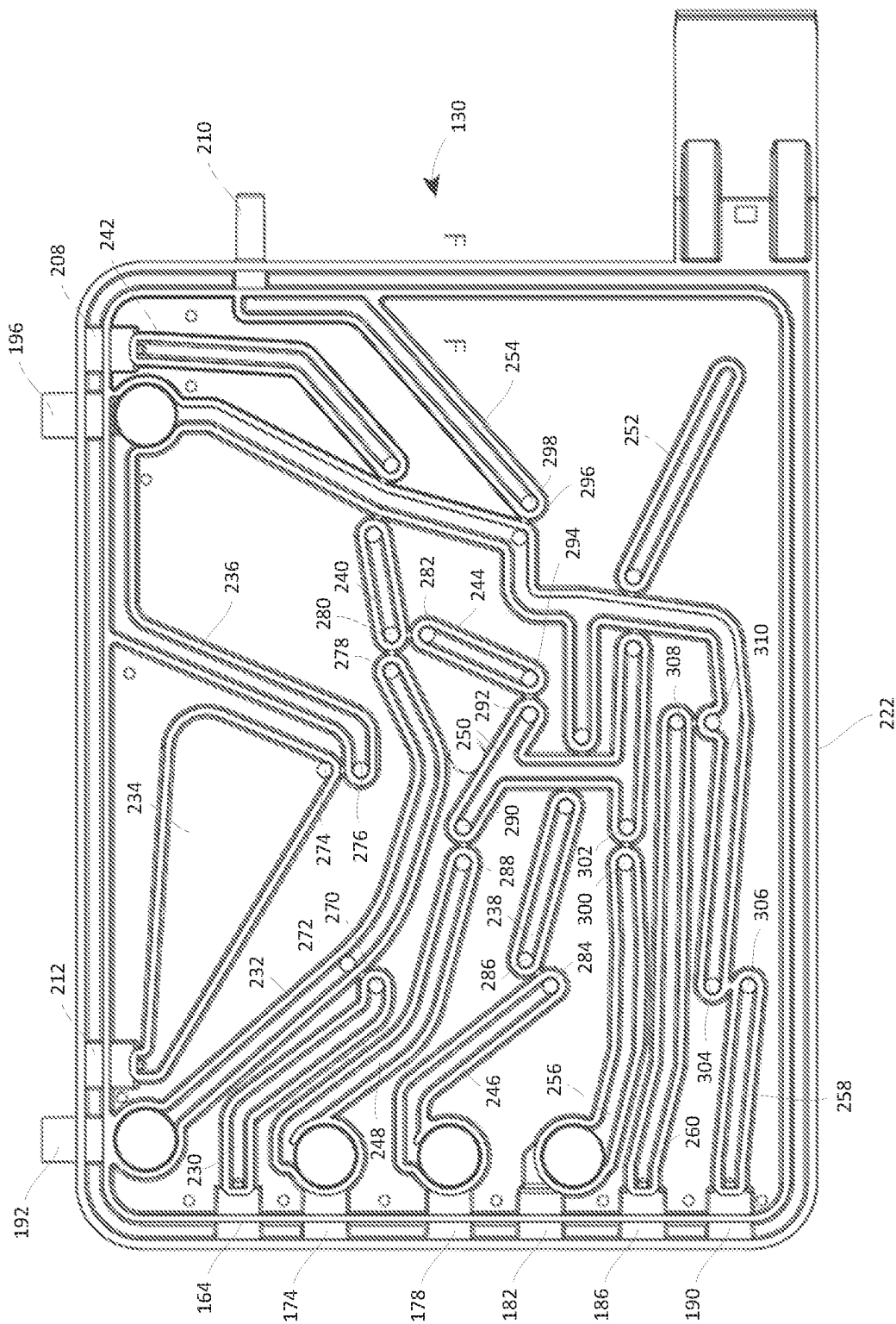
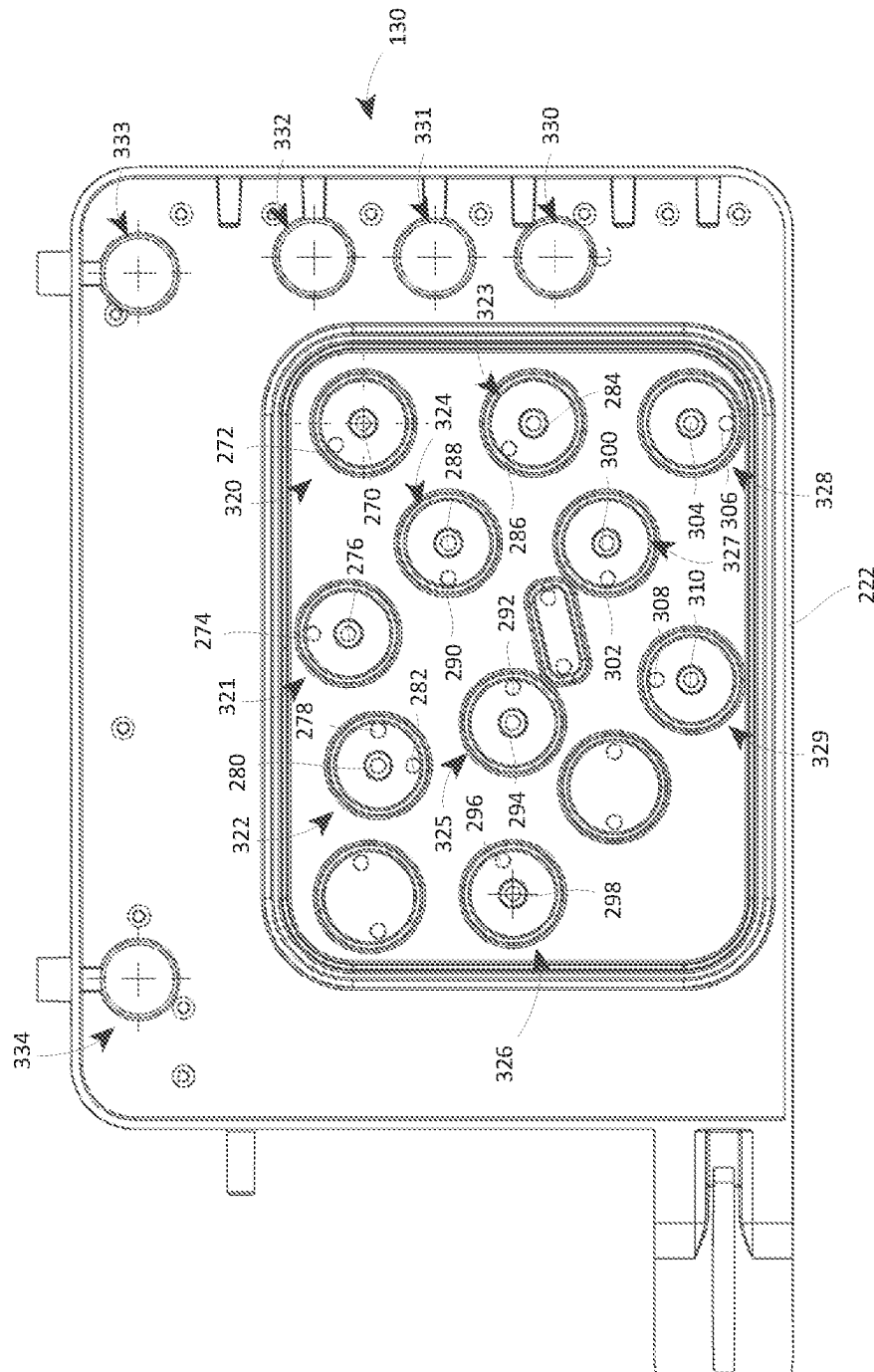


FIG. 4



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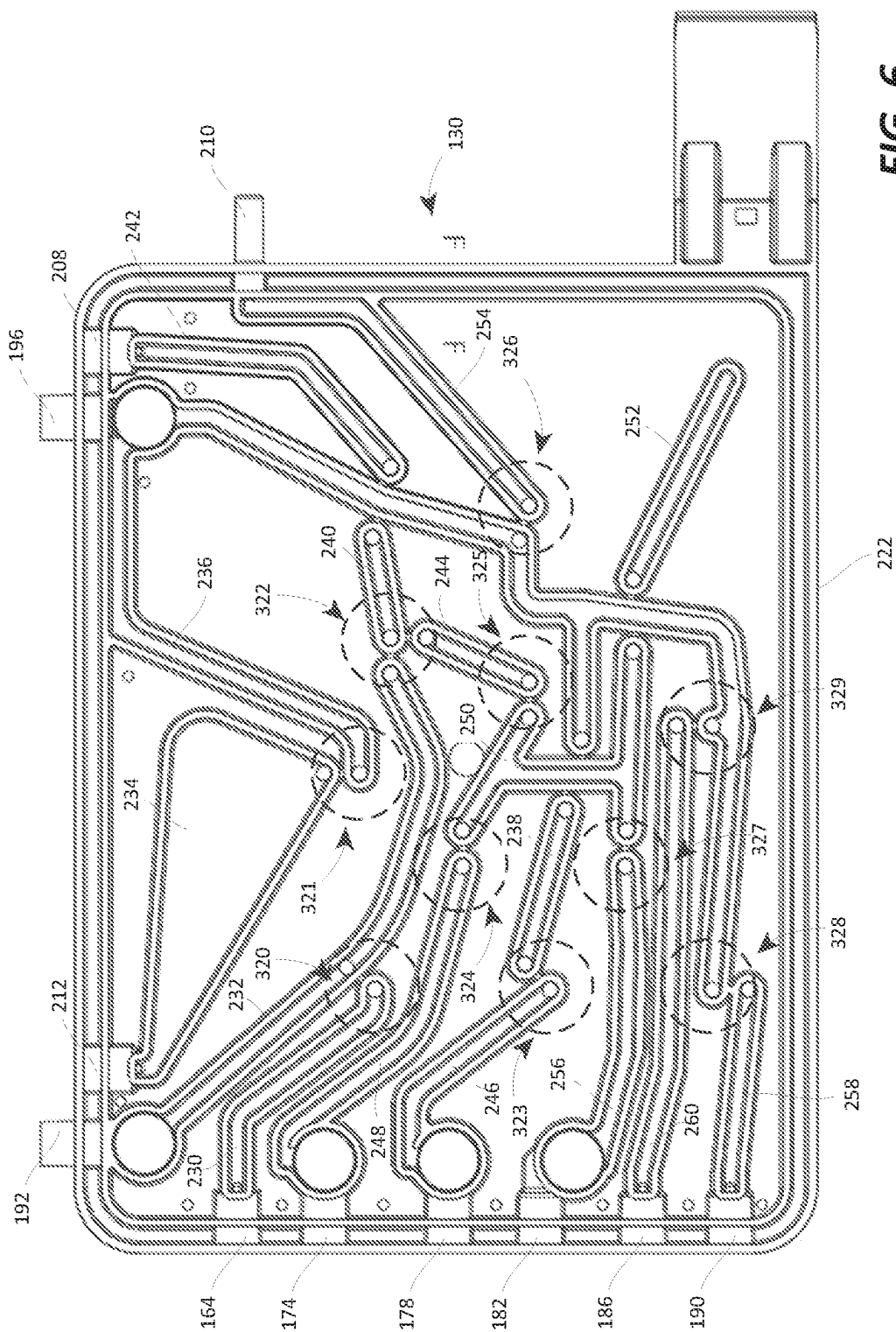


FIG. 6

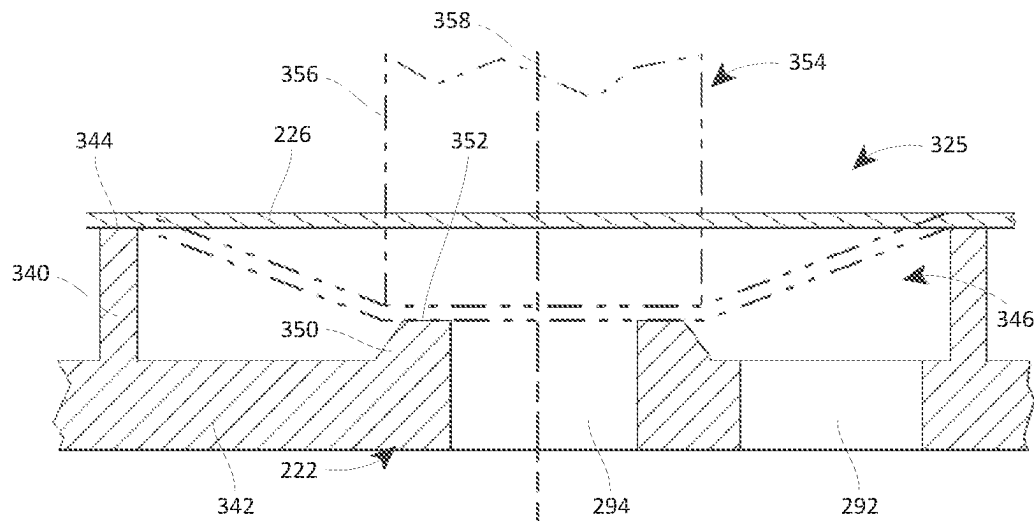


FIG. 7

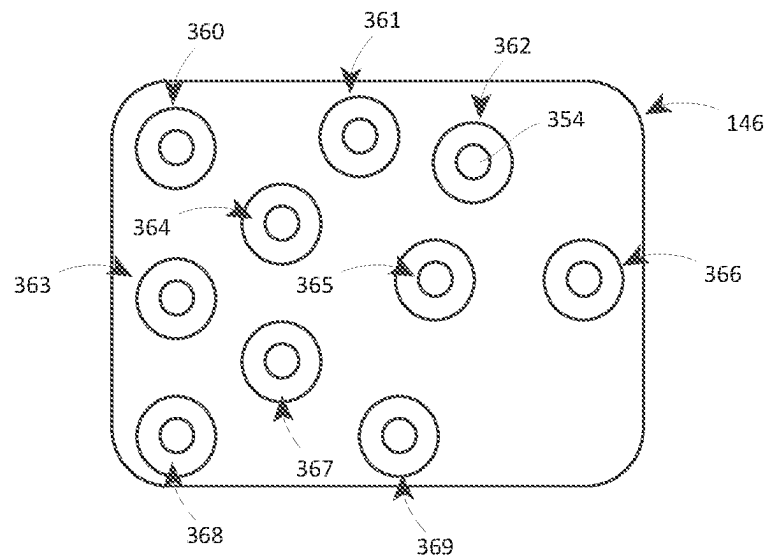


FIG. 8

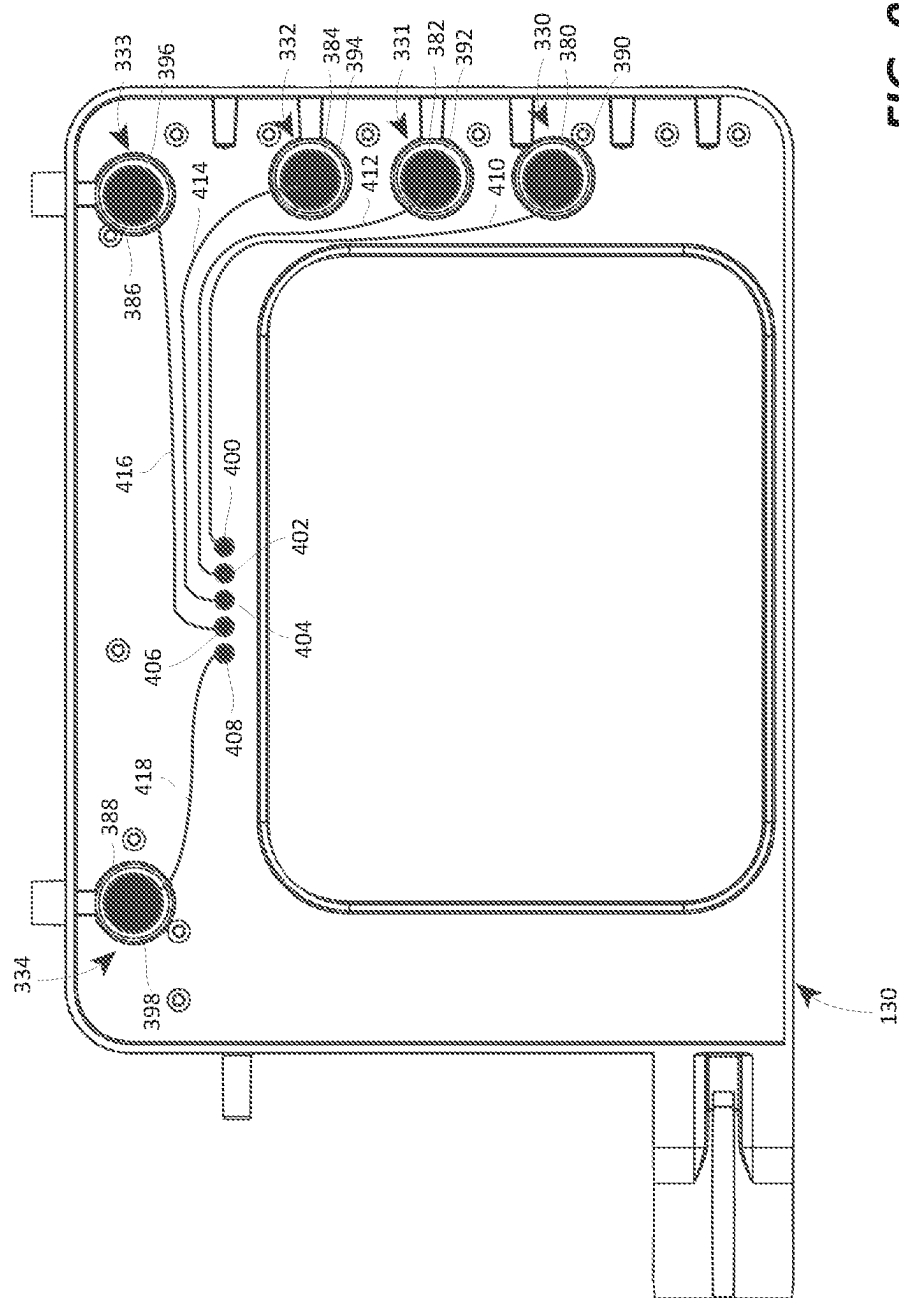


FIG. 9

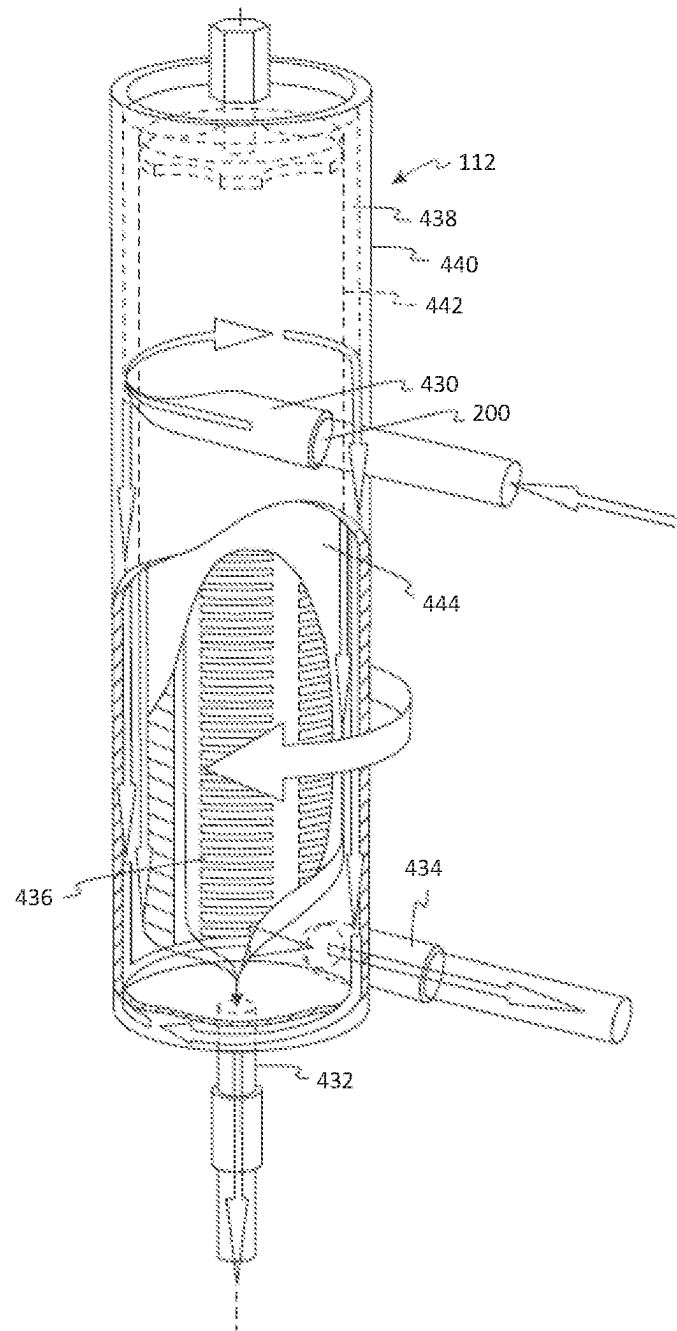
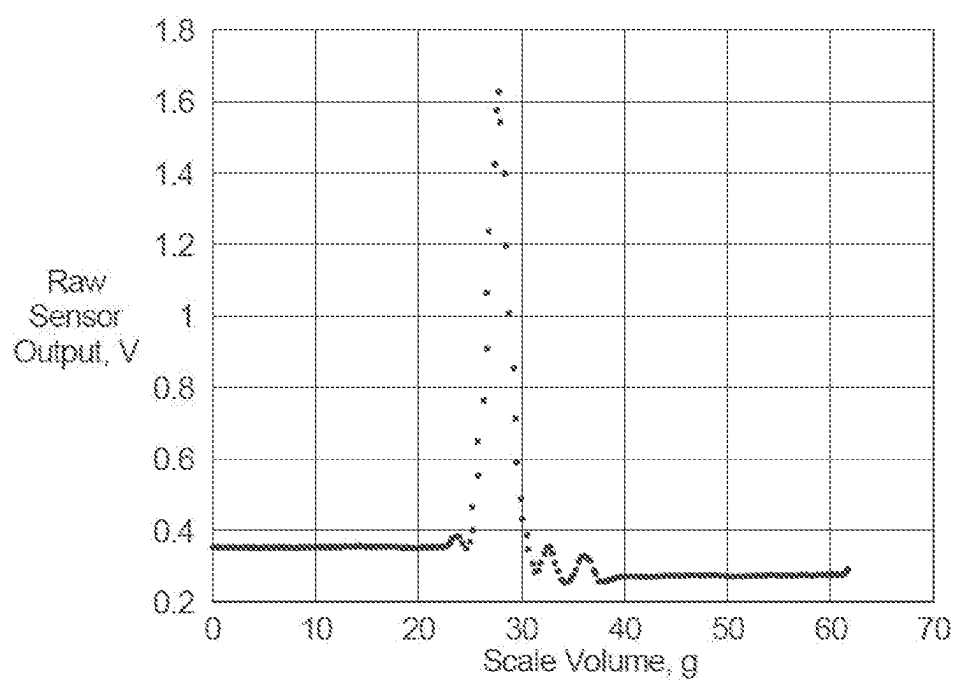
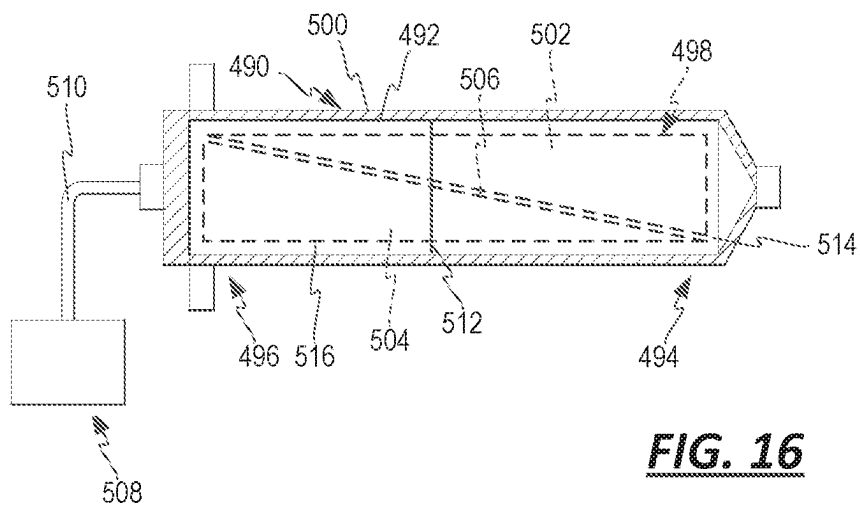
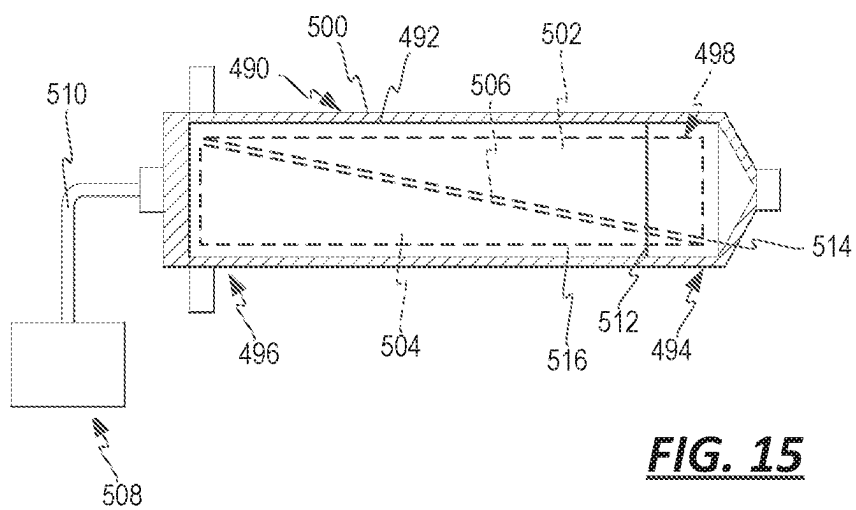
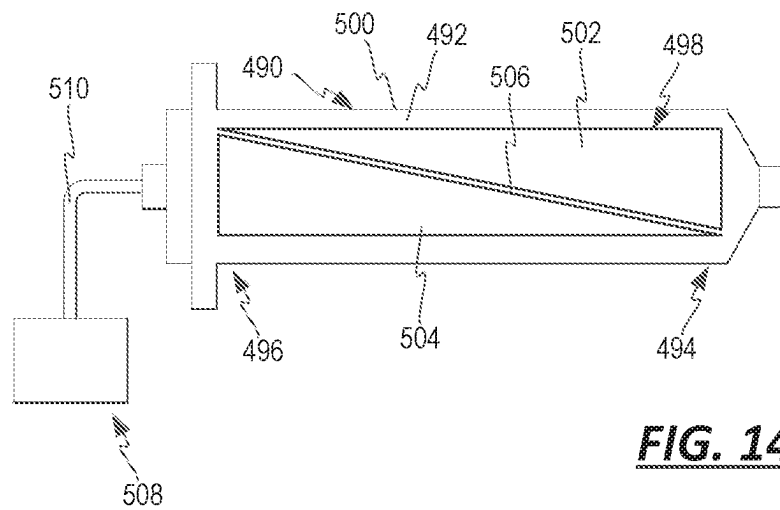
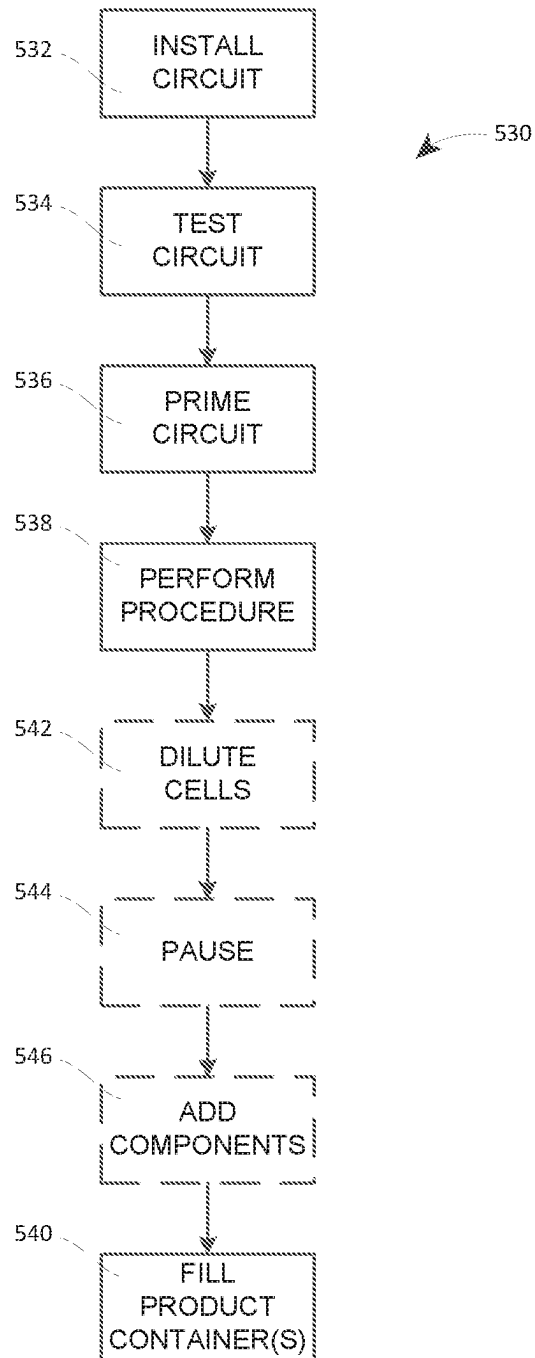


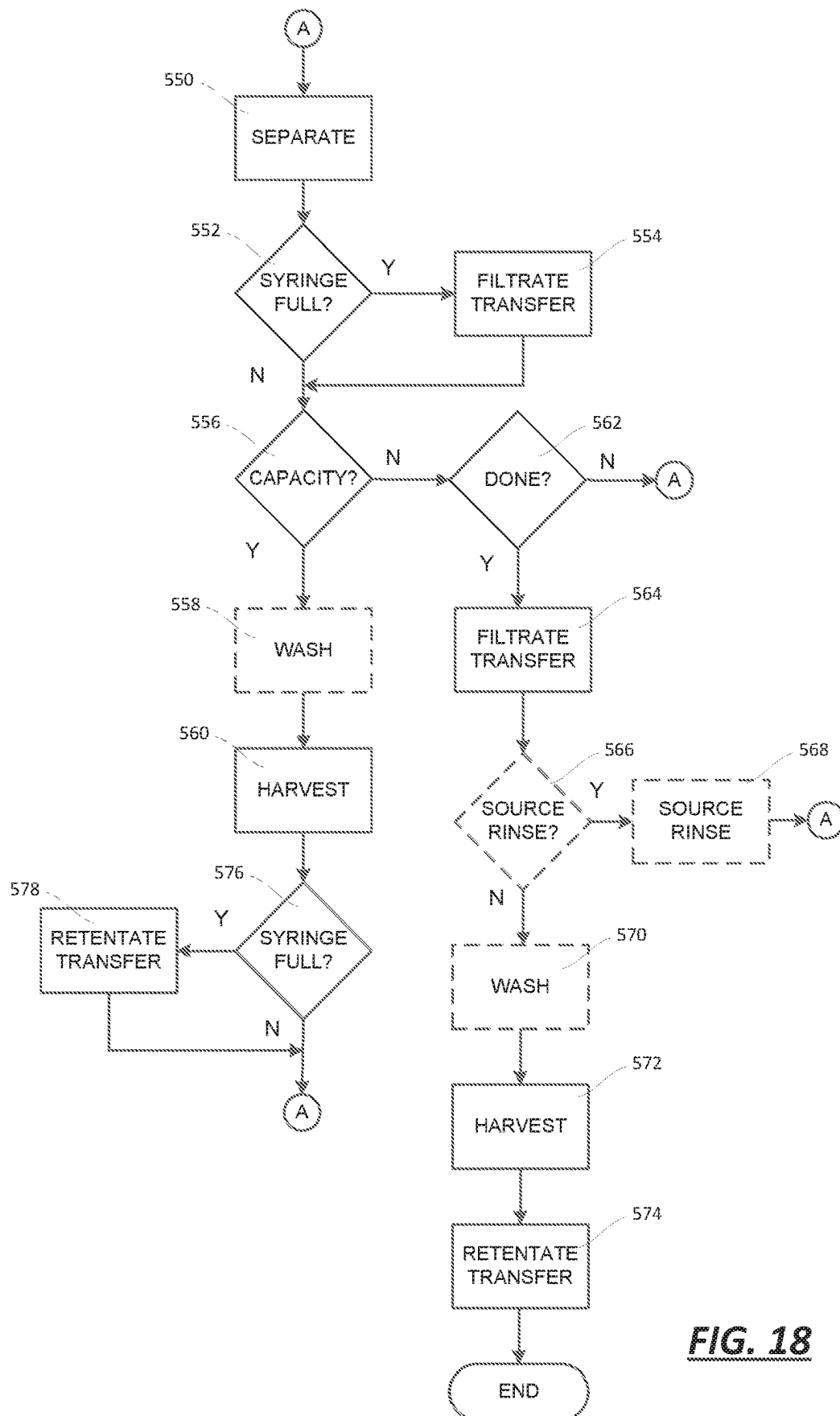
FIG. 10

FIG. 12

**FIG. 13**



***FIG. 17***

**FIG. 18**

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SMALL VOLUME PROCESSING SYSTEMS AND METHODS WITH CAPACITIVE SENSING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Patent App. No. 62/925,303, filed Oct. 24, 2019, which is expressly incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present disclosure is generally directed to systems and methods for processing (e.g., washing) suspensions of cells. More particularly, the present disclosure is directed to systems and methods for processing small volumes of cells using a disposable fluid circuit and a reusable processing machine or hardware to generate a cell product.

BACKGROUND

A number of well-known therapies are currently practiced in which a targeted cellular blood component (e.g., red blood cells, white blood cells, and platelets) is separated from whole blood and stored for later infusion to a patient. The targeted cell product (e.g., red blood cells, white blood cells, or platelets) may be in a suspension that includes plasma and/or some other supernatant. As such, it is sometimes desirable to “wash” the cellular suspension (typically with a physiologic buffer) to remove the plasma/supernatant, as well as any non-target cellular material, prior to reinfusion.

Systems and methods for cell washing are exemplified by US Pub. Nos. 2013/0341291, 2013/0092630, and 2014/0199680, each of which is incorporated herein by reference. Each of these published applications discloses cell washing methods utilizing disposable fluid circuits including a spinning membrane separator and a reusable processing machine. Such machines include peristaltic pumps and pinch valves that act on the tubing of the fluid circuit to direct flow within the fluid circuit.

The fluid circuits in the published applications listed above have a relatively large internal volume, and thus require relatively large volumes of wash or flush media to clear processed fluid through the fluid circuit. While such systems and fluid circuits are capable of washing and reducing the volume of the targeted cell component into final volumes of ranging from approximately 50 mL to 5,000 mL, there are instances in which smaller final volumes (e.g., 10 mL) are desired, such as when processing single-dose quantities of mononuclear cell products. Thus, it would be desirable to provide systems and methods for processing (e.g., concentrating or washing) small volumes of cellular suspensions.

SUMMARY

In an aspect, a fluid processing system includes a disposable fluid circuit and reusable hardware configured to accept the disposable fluid circuit. The disposable fluid circuit includes a separation chamber and a flow control cassette. The flow control cassette includes a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one interface sensor chamber in fluid communication with at least one of the plurality of separate channels, the at least one interface sensor chamber defined at least in part by a wall, and at least one capacitive

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sensor disposed on the wall of the at least one interface sensor chamber. The reusable hardware includes a separator connected to the separation chamber, a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor, and at least one controller coupled to the separator, the at least one actuator and the capacitive sensor via the coupling, the controller configured to selectively operate the separator and the at least one actuator to provide a procedure according to a protocol.

In another aspect, a fluid processing system includes a disposable fluid circuit and reusable hardware configured to accept the disposable fluid circuit. The disposable fluid circuit includes a separation chamber, at least one plungerless syringe, the syringe including a wall defining a barrel having a first end and a second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the syringe, and a flow control cassette including a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one of the plurality of separate channels connected to the first end of the barrel of the at least one plungerless syringe. The reusable hardware includes a separator connected to the separation chamber, at least one syringe pump, the second end of the barrel of the at least one syringe coupled to the at least one syringe pump, the at least one syringe pump configured to draw fluid into and push fluid from the at least one syringe, a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor, and at least one controller coupled to the separator, the at least one syringe pump and the control cassette interface, the controller configured to selectively operate the separator, the at least one syringe pump and the interface to provide a procedure according to a protocol.

In a further aspect, a fluid processing system includes a disposable fluid circuit and reusable hardware configured to accept the disposable fluid circuit. The disposable fluid circuit includes a separation chamber; at least one plungerless syringe, the syringe including a wall defining a barrel having a first end and a second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the syringe, and a flow control cassette. The flow control cassette includes a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one of the plurality of separate channels connected to the first end of the barrel of the at least one plungerless syringe, at least one interface sensor chamber in fluid communication with at least one of the plurality of separate channels, the at least one interface sensor chamber defined at least in part by a wall, and at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber. The reusable hardware includes a separator connected to the separation chamber, at least one syringe pump, the second end of the barrel of the at least one syringe coupled to the at least one syringe pump, the at least one syringe pump configured to draw fluid into and push fluid from the at least one syringe, a coupling connected to the at least one capacitive sensor disposed on an outer surface of the syringe, a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber, and at least one controller coupled to the separator, the at least one syringe pump and the control cassette interface, the control-

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ler configured to selectively operate the separator, the at least one syringe pump and the interface to provide a procedure according to a protocol.

DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a system for processing (e.g., concentrating or washing) small volumes of cellular suspensions including a disposable fluid circuit and a reusable processing machine or hardware.

FIG. 2 is a plan view of an embodiment of a disposable fluid circuit for use in the system of FIG. 1.

FIG. 3 is a block diagram of an embodiment of the reusable processing machine or hardware of FIG. 1.

FIG. 4 is a side view of an embodiment of a flow control cassette for use in the disposable fluid circuit of FIG. 2, with one of the sidewalls removed for ease of illustration.

FIG. 5 is a side view of the embodiment of the flow control cassette of FIG. 4, illustrating the opposite side of the flow control cassette.

FIG. 6 is a side view of the embodiment of the flow control cassette of FIG. 4, with the location of the selectable junctions, or clamps, from the opposite side of the cassette illustrated relative to the channels and apertures illustrated in FIG. 4.

FIG. 7 is a cross-sectional view of one of the clamps of the flow control cassette being manipulated by an actuator.

FIG. 8 is a plan view of an embodiment of a flow control interface that is associated with the flow control cassette, and which is configured to operate the selectable junctions, or clamps, of the flow control cassette.

FIG. 9 is a side view of another embodiment of a flow control cassette for use in the disposable fluid circuit of FIG. 2, the cassette incorporating capacitive sensors for use with air sensor chambers;

FIG. 10 is a perspective view of a separation/washing device using a spinning membrane.

FIG. 11 is a cross-sectional view of an embodiment of a syringe/syringe pump as may be used as part of the system of FIG. 1, with a piston head assembly in a first position.

FIG. 12 is a cross-sectional view of the syringe/syringe pump of FIG. 11, with a piston head assembly in a second position.

FIG. 13 is a chart of the signal response of one of the infrared detectors or sensors of the syringe/syringe pump of FIG. 11.

FIG. 14 is a side view of another embodiment of a syringe/syringe pump as may be used as part of the system of FIG. 1, with a capacitive sensor applied to an outer surface of the syringe.

FIG. 15 is a cross-sectional view of the syringe/syringe pump of FIG. 14, with an air/fluid interface in a first position.

FIG. 16 is a cross-sectional view of the syringe/syringe pump of FIG. 14, with an air/fluid interface in a second position.

FIG. 17 is a flowchart illustrating an embodiment of a method of operating the system of FIG. 1.

FIG. 18 is a flowchart illustrating an embodiment of a procedure that may be performed according to a protocol using the system 100.

DETAILED DESCRIPTION

A more detailed description of the systems and methods in accordance with the present disclosure is set forth below. It should be understood that the description below of specific

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devices and methods is intended to be exemplary, and not exhaustive of all possible variations or applications. Thus, the scope of the disclosure is not intended to be limiting, and should be understood to encompass variations or embodiments that would occur to persons of ordinary skill.

Turning first to FIG. 1, an embodiment of a system 100 for processing fluids, such as cell suspensions (e.g., cell washing), is illustrated, the system 100 including a disposable fluid circuit (also referred to as a set or kit) 102 and a reusable processing machine, or hardware, 104.

As seen in FIGS. 1 and 2, the disposable fluid circuit 102 is connectable to a source container 110 of fluid, in particular biological fluid. The disposable fluid circuit 102 includes a separator chamber, in this case defined by a spinning membrane separator 112 that is used to process the fluid received from the source container 110, and to direct a portion of that fluid into one of more product containers 114. These containers may be in the form of flexible bags according to the illustrated embodiment. The flow of fluid from the source container 110, through the spinning membrane separator 112, and to the one or more product containers 114 is achieved through the use of first and second syringes 116, 118, which are in fluid communication with the source container 110, the spinning membrane separator (or spinning membrane for short) 112, and the one or more product containers 114. The syringes 116, 118 also may be in fluid communication with a number of other containers 120, 122, 124, 126, 128 (container 124 being schematically represented in FIG. 2 only).

The flow of the fluid between the containers 110, 114, 120, 122, 124, 126, 128, the spinning membrane 112, and the syringes 116, 118 is controlled using a flow control cassette 130, which cassette 130 may be connected to each of the foregoing by tubing, or lines. In addition, the cassette 130 may include internal flow paths that are defined in part by a plurality of separate channels or passages, which in turn may be contained within the structure (e.g., housing) of the cassette 130. The channels may be defined by the structure (e.g., housing) of the cassette 130, as illustrated, or may be defined by tubing or lines disposed within the cassette, see U.S. Pub. No. 2017/0204371, which is incorporated herein by reference. The channels may be connected at a plurality of selectable junctions, which may control the flow of fluid from one channel to another. These selectable junctions may also be referred to as valves, valve stations, or clamps, because, as illustrated, the selectable junctions provide controlled access between the channels. The cassette 130 may also include sensor stations, by which sensors may be associated with the flow paths within the cassette 130 to determine characteristics of the flow therein, such as pressure, presence of air and/or fluid, or optical properties. Preferably, the length of each of the lines and channels is kept as short as possible to further minimize the internal volume of the fluid circuit 102.

As illustrated in FIG. 2, the spinning membrane 112 and the syringes 116, 118 may be integrally formed as part of (i.e., as one piece with) the cassette 130, so as to further reduce the tubing volume associated with the kit 102. According to other embodiments, the spinning membrane 112 and/or the syringes 116, 118 may be attached to the remainder of the fluid circuit 102 at the time of use, as may be the case with one or more of the containers 110, 114, 120, 122, 124, 126, 128. Again, as illustrated in FIG. 2, the container 120 and container 128 are integrally formed with the cassette 130.

As seen in FIGS. 1 and 3, the reusable hardware component (or reusable hardware for short) 104 includes a drive

140 for the spinning membrane separator 112, a syringe pump 142, 144 for each syringe 116, 118, and a control cassette interface 146 that is associated with the flow control cassette 130 when the fluid circuit 102 is disposed on the hardware 104 (e.g., is mounted on the hardware 104). As will be explained in detail below, the cassette interface 146 includes actuators and sensors that are associated with the clamps and sensor stations of the flow control cassette 130, and are configured to operate the clamps or sense characteristics of the fluid, respectively.

The reusable hardware 104 also includes a controller 150 that is configured to control operation of the system 100, for example using a method of operation as is explained below relative to FIGS. 17 and 18. The controller 150 may include a microprocessor 152 (which, in fact may include multiple physical and/or virtual processors). According to other embodiments, the controller 150 may include one or more electrical circuits designed to carry out the actions described herein. In fact, the controller 150 may include a microprocessor 152 and other circuits or circuitry. In addition, the controller 150 may include one or more memories 154. The instructions by which the microprocessor 150 is programmed may be stored on the one or more memories 154 associated with the microprocessor 150, which memory/memories 154 may include one or more tangible non-transitory computer readable memories, having computer executable instructions stored thereon, which when executed by the microprocessor 152, may cause the microprocessor 152 to carry out one or more actions as described below.

The controller 150 may be coupled (i.e., directly or indirectly connected) to the equipment of the reusable hardware 104, such as the spinning membrane drive 140, the first syringe pump 142, the second syringe pump 144, and the cassette interface 146. The controller 150 may operate each of these devices, each of which may be an assembly of other devices or equipment, to cause the fluid to flow through the fluid circuit 102 associated with the hardware 104, for example to cause fluid to flow from the source container 110, through the spinning membrane 112, and eventually into the product container(s) 114. For example, the controller 150 may be programmed to perform a process or procedure according to a protocol, such as to wash particular cells contained in the fluid within the source container 110, before they are distributed into one or more of the product containers 114. The controller 150 may be programmed to perform other actions as well, such as to test the fluid circuit 102, to prime the fluid circuit 102, to rinse parts of the circuit 102 after the wash has been performed, to add other components to the cell-containing fluid before that fluid is distributed to the product containers 114, and to distribute the cell-containing fluid into the product containers 114.

Having thus described the structure and operation of the system 100, including the fluid circuit 102 and reusable hardware 104, in general terms, the details of each of the systems is now discussed, starting with the fluid circuit 102.

As mentioned above, the flow of fluids through the fluid circuit 102 is controlled through the flow control cassette 130. While other embodiments may involve fluid circuits 102 where some of the fluid does not pass through the cassette 130, according to the illustrated embodiment, the fluid flows between the containers 110, 114, 120, 122, 124, 126, 128, the spinning membrane 112, and the syringes 116, 118 via the cassette 130. As mentioned above, each of the containers 110, 114, 120, 122, 124, 126, 128, the spinning membrane 112, and the syringes 116, 118 is connected to the cassette through the use of medical grade tubing, or lines.

With reference to FIGS. 1 and 2, the container 120 used to receive the filtrate of the spinning membrane 112 and other fluids is connected via a line 162 to a filtrate container port 164 formed on the cassette 130. The first and second containers 122, 124, used to contain wash media as may be used during the method of operation of the system 100, each may be connected to a line 166, 168 that are connected at a first end to the containers 122, 124, and at a second end to a Y-junction 170. The Y-junction 170 is, in turn, connected via a line 172 to a wash container port 174. The container 126, which may contain a cryopreservation agent (CPA) according to one embodiment, is connected via a line 176 to a port 178. The source container 110 may be connected via a line 180 to a source container port 182. Further, a secondary container 128 is connected via a line 184 to a port 186, and the product container(s) 114 is/are connected via a line 188 to a product container port 190.

As is reflected in the illustrated embodiment, certain of the containers may be formed integrally with the fluid circuit 102, while other containers may be attached at the time of operation. For example, filtrate container 120 and the secondary container 128 are formed integrally with their respective lines 162, 188.

On the other hand, lines 166, 168, 176, 180, 188 may be formed with an attachment site (such as an end formed to be sealed to the container or with a connector, such as a luer lock connector, attached thereto) to connect to the containers 110, 122, 124, 126, 114 at the time of use.

The syringe 116 may be connected via a line 192 to a port 194, and the syringe 118 may be connected via a line 196 to a port 198. In a similar fashion, the spinning membrane 112 may be connected at an inlet of the spinning membrane 112 by a line 200 to an inlet port 202, and at a first outlet via a line 204 to a first outlet port 208 and at a second outlet via a line 206 to a second outlet port 210. In addition, an air vent port 212 is provided, and the air vent port 212 is connected to a filter 214 via a line 216. Because of the proximity of the spinning membrane 112, the syringes 116, 118, and the filter 214 to the cassette 130, one or more of the lines 192, 196, 200, 204, 208, 216 may be defined by portions of cassette 130 itself.

According to the illustrated embodiment, and as seen in FIGS. 2 and 4-6, the cassette 130 includes a housing 220 defined by a frame 222 to which side walls 224, 226 are attached (see FIGS. 2 and 7). The walls 224, 226 may be attached about the periphery of the frame 222, as well as along structures of the frame that define the channels, clamps, and sensor stations discussed above. The walls 224, 226 may be attached through the use of joining techniques, such as ultrasonic welding, or may be attached by holding the wall 224, 226 and the frame 222 in contact with each other through the application of force.

A negative pressure may be drawn on the side wall 224 of the cassette 130. Drawing a negative pressure on the wall 224 of the cassette 130 is believed to prevent the collapse of the channels defined within the housing 220. This is particularly important in a system that uses syringes 116, 118 and syringe pumps 142, 144 in that the syringe pumps operate, at least in part, by drawing negative pressures within the fluid paths defined, at least in part, by the channels. The application of negative pressure to the wall 224 of the cassette 130 compensates, at least in part, for the negative pressures drawn within the fluid paths.

Turning next to FIG. 4, it will be noted that the frame 222 defines the afore-mentioned plurality of separate and distinct channels, which channels may be connected to one of the ports discussed above. The channels may also have one or

more apertures disposed at locations along the lengths of the channel. These apertures may be used to connect the channels, via the clamps or sensor stations, for example, to other channels. Together, the channels may define flow paths (or fluid paths, or fluid flow paths) between the containers **110**, **114**, **122**, **124**, **126**, **128**, syringes **116**, **118**, and the spinning membrane **112**.

Starting at the left-hand side of the cassette **130**, a channel **230** is connected to the port **164**, and includes an aperture **270**. A channel **232** is connected to port **192**, and includes apertures **272**, **278**. A channel **234** is connected to port **212**, and includes aperture **274**. A channel **236** is connected to port **196**, and includes apertures **276**, **296**, **304**, **310**. A channel **238** includes an aperture **286**, while a channel **240** includes an aperture **280**. A channel **242** is connected to port **208**, and is connected to channel **240** via a station and unnumbered apertures of channels **240**, **242**; in a similar fashion, channel **238** is connected to channel **236** via a station and unnumbered apertures of channels **236**, **238**.

Towards the middle of the cassette **130**, a channel **244** includes apertures **282**, **294**. A channel **246** is attached to port **178**, and includes aperture **284**. A channel **248** is attached to port **174**, and includes aperture **288**. A channel **250** includes apertures **290**, **292**, **302**. A channel **252** is connected to the inlet port **202** of the spinning membrane **112**, and is connected to the channel **250** via a station and unnumbered apertures of channels **250**, **252**.

At the right-hand side of the cassette **130**, a channel **254** is connected to port **210**, and includes aperture **298**. On the other hand, at the right-hand side of the page, a channel **256** is connected to port **182** and includes an aperture **300**. A channel **258** is connected to port **190**, and includes an aperture **306**. Finally, a channel **260** is connected to port **186**, and includes an aperture **308**.

As seen in FIG. 5, each of the apertures included in the channels **236-260** is associated with one or more of the other apertures. In most instances, each aperture is associated with one of the other apertures; in one instance, three apertures are associated with each other. Each grouping of two or more apertures is associated with a chamber on the reverse side of the cassette **130** from that illustrated in FIG. 4, which chamber then defines one of the clamps.

In particular, apertures **270**, **272** are grouped, and define in part a selectable junction or clamp **320**, while apertures **274**, **276** are grouped, and define in part a clamp **321**. The apertures **278**, **280**, **282** are grouped, and define in part a clamp **322**. The apertures **284**, **286** define in part a clamp **323**, the apertures **288**, **290** define in part a clamp **324**, the apertures **292**, **294** define in part a clamp **325**, and the apertures **296**, **298** define in part a clamp **326**. Finally, the apertures **300**, **302** define in part a clamp **327**, the apertures **304**, **306** define in part a clamp **328**, and the apertures **308**, **310** define in part a clamp **329**. The clamps **320-329** are also shown in dashed line in FIG. 6 with the markings of the apertures removed, for ease of illustration relative to the associations of the clamps **320-329** with the channels **230-260**.

As mentioned above, each of the groupings of apertures is associated with a chamber, which chamber and the features thereof further define one of the clamps **320-329**. An exemplary clamp (for example, clamp **325**) is illustrated in larger scale in FIG. 7 so that the cooperation of the structures of the chamber may be visualized (the structures of the corresponding channels have been omitted for ease of illustration). While the clamp illustrated in FIG. 7 has only two apertures in cross-section, this structure also is applicable to those clamps that have more than two apertures.

The clamp illustrated includes a chamber wall **340** that is formed as part of the frame **222**, and extends from a frame wall **342**. The chamber wall **240** encloses a circular region as viewed in FIG. 5, and thus may also be described as a circumferential or peripheral wall. The side wall **226**, which may be made of a flexible material, is attached to an edge **344** of the chamber wall **340**, and with the chamber wall **340** and the frame wall **342** define an enclosed region or space **346**. The apertures **292**, **294** pass through the frame wall **342**, and one of the apertures **294** has a rim or flange **350** disposed about its circumference or periphery. The distance of an edge **352** of the rim **350** from the frame wall **342** is not as great as the distance of the edge **344** from the frame wall **342**.

As illustrated in dashed line in FIG. 7, a portion of the wall **226**, also referred to as a deflectable surface, may be brought into contact with the edge **352** of the rim **350** to cover the aperture **294** so that fluid cannot flow between the aperture **294** and the volume or space **346**. At the same time, the fluid flow between the aperture **292** and the space **346** may remain unobstructed because the deflectable surface does not cover the aperture **292**. However, by closing the aperture **294**, fluid flow may be interrupted along the fluid path defined by the channels **244**, **250** associated with the apertures **294**, **292**, respectively. The wall **226** may be selectively deflected to abut the rim **350** through the use of an actuator **354** coupled to the controller **150**, which actuator **354** may be defined in part by a shaft **356** that moves along an axis **358** (for example, where the shaft **356** is part of an electronic linear actuator). When the shaft **356** is advanced in the direction of the wall **342**, the shaft **356** deflects the wall **226** to abut the edge **352** and close the aperture **294**. When the shaft **356** is withdrawn away from the wall **342**, the wall **226** moves away from the edge **352** and the aperture **294** is open and in fluid communication with the space **346**.

As illustrated in FIG. 8, the cassette interface **146** may include a plurality of actuators **354**, each within a space that is intended to be aligned with one of the clamps **320-329**. The actuators **354** each may define one of a plurality of actuator stations **360-369** that corresponds to a respective one of the clamps **320-329**. Each of the actuators stations **360-369** may be coupled to the controller **150**, as illustrated in FIG. 3, and the controller **150** may control the movement of the actuators **354** in the direction of and away from the cassette **130** when the cassette **130** is disposed or mounted on the cassette interface **146**. The controller **150** may operate the actuators **354** in conjunction with the desired process.

According to an alternative embodiment of the cassette, where the channels are defined not by structures of the housing, but by tubing or lines disposed within the housing, the clamps may be defined by pinch valves. See U.S. Pub. No. 2017/0204371, which is incorporated herein by reference.

The cassette **130** may also include a number of air sensor chambers **330-334** disposed at points along the periphery of the frame **222**. See FIG. 5. The cassette **130** is to be used with air sensors that are associated with each of the air chambers **330-334** such that it is not necessary that the emitter and detector be disposed on opposite sides of the frame **222**. Instead, the emitter and detector can be disposed on the same side of the frame **222**, providing a so-called single-sided air sensor. This may be beneficial because there is no need to provide a door to close over the cassette **130**, the door having either an emitter or a detector mounted thereon, as would be the case with a pass-through sensor where the emitter and detector must be disposed on opposite sides of the cassette **130**.

The single-sided air sensors may be in the form of an ultrasonic sensor that emits controlled, timed pulses of ultrasonic energy into the chamber 330-334 and senses the response time of the "echo" of the emitted energy. The ultrasonic sensors may be part of the reusable hardware 104 of the system 100, and thus may be mounted opposite the cassette 130 with the cassette interface 146. The ultrasonic sensors may each be spring-biased (e.g., combined with a spring-loaded insert), such that the sensors will come into contact with a wall of each of the chambers 330-334 when the cassette 130 is mounted to the hardware 104, and in particular the cassette interface 146. The echo time is believed to change when the liquid enters the chamber.

As an alternative, each of the chambers may have a window that permits an optical sensor to be used therewith, the window being translucent at least to light of a wavelength emitted by a light emitter associated with the sensor. A single-sided (reflectance-based) optical sensor may be used to determine other things than the presence of an air/fluid interface, such as cell concentrations in the fluid as well.

A further alternative is illustrated in FIG. 9. According to this embodiment (in which the cassette 130 is otherwise structurally identical to the cassette 130 of FIGS. 2 and 4-6), the single-sided sensors 380, 382, 384, 386, 388 are capacitive sensors. Instead of being part of the reusable hardware 104 of the system 100, the sensors 380, 382, 384, 386, 388 are attached to the cassette 130, and are part of the disposable fluid circuit 102 instead. As illustrated in FIG. 9, the sensors 380, 382, 384, 386, 388 are disposed on a wall 390, 392, 394, 396, 398 of the chambers 330-334. The sensors 380, 382, 384, 386, 388 may be attached to an outer surface of the respective wall 390, 392, 394, 396, 398 of the respective chamber 330-334 with an adhesive, for example.

Each of the sensors 380, 382, 384, 386, 388 is connected to one of a plurality of connector pads 400, 402, 404, 406, 408 that are disposed on the cassette 130, in particular on an outer surface of the cassette 130. The connector pads 400, 402, 404, 406, 408 may be connected to the sensors 380, 382, 384, 386, 388 by a lead or trace 410, 412, 414, 416, 418 that also may be formed on the outer surface of the cassette 130. The connector pads 400, 402, 404, 406, 408 are configured to be contacted by a plurality of connectors mounted on the reusable hardware 104 of the system 100. For example, the hardware 104 may include a plurality of spring-biased or spring-loaded connectors (e.g., pogo pins) that are configured to contact the connector pads 400, 402, 404, 406, 408.

The capacitive sensors 380, 382, 384, 386, 388 may be in the form of a multi-layer electrical elements, such as a flexible, or flex, circuit. The sensors 380, 382, 384, 386, 388 may monitor the electrical capacity of a known area and sense distortions within the electrostatic field in that area when other materials (other than air) are proximate to the area. In particular, the sensor may include two elements separated by a gap from each other, creating a structure having a capacitance that varies when either air or a fluid (e.g., fluid within the chamber 330-334) is proximate to the gap. In this fashion, the sensor 380, 382, 384, 386, 388 may determine whether air or fluid is in the chamber 330-334.

An embodiment of the cassette 130 including capacitive sensors 380, 382, 384, 386, 388 to be used to determine the presence or absence of fluid in the chambers 330-334 may have one or more of the following advantages. Direct application of the sensors 380, 382, 384, 386, 388 to the cassette 130 limits or eliminates the effect that variations in dimensional tolerances may have on the sensor's detection

abilities where the sensors must instead be brought into contact with the cassette 130, and particularly the chambers 330-334. Additionally, the spatial distribution of the chambers 330-334 over the cassette 130 means that the tolerances must be considered over a large area of the surface of the cassette 130 if the sensors must be brought into contact with the chambers 330-334. By comparison, the connector pads 400, 402, 404, 406, 408 may be localized to a specific region of the cassette 130. Further, instead of having to configure the entire sensor to be adjustable to address tolerances in the manufacture of the cassette 130, the embodiment of FIG. 9 permits the tolerances to be addressed through the design of the connector, which may be more easily achieved through the use of conventional spring-biased or spring-loaded connectors or other traditional connector technologies.

While the afore-mentioned use of flexible capacitive sensors has been made with reference to sensing air-fluid interfaces in a cassette, such as the embodiments of the cassette 130 illustrated herein, it will be recognized that these capacitive sensors may be used with other equipment, and particularly medical equipment, where an air-fluid interface must be determined relative to fluid flowing in a disposable circuit that is associated with reusable hardware. For example, the technology may be used other separation technologies, such as an AMICUS® separator or an Autopheresis C® system, both of which are available from Fresenius Kabi USA, Lake Zurich, Ill.

In addition to the cooperation between the cassette 130 and the cassette interface 146, the disposable fluid circuit 102 and the reusable hardware 104 cooperates in other ways as well.

FIG. 10 illustrates additional details of the spinning membrane separator 112, for example. Preferably, spinning membrane 112 is a spinning membrane separator of the type described in U.S. Pat. Nos. 5,194,145 and 5,053,121, U.S. Provisional Patent Application No. 61/451,903 and International Pub. No. WO 2012/125470, which are incorporated herein by reference in their entirety. As discussed above, the spinning membrane separator 112 has one inlet 430 at least two outlet ports 432, 434. The outlet 432 of spinning membrane 112 receives the waste from the wash (i.e., a non-cellular component of the cellular suspension and wash medium from the spinning membrane separator) and is connected to line 204. The spinning membrane 112 preferably includes a second outlet 434 that is connected to line 206 and receives the desired biological cell/fluid product (e.g., washed cells).

FIGS. 11 and 12 illustrate an embodiment of a syringe pump that may be used with either the first or the second syringe 116, 118 and as either the first and/or second syringe pump 142, 144.

The syringe pump is configured to use a syringe 450 with a syringe barrel 452 (which may be made of cyclic olefin copolymer, or other materials such as may be inert, optically clear) and a piston or plunger head assembly 454. The piston head assembly 454 is moveable (translatable) between a first end 456 and a second end 458 of the barrel 452.

The piston head assembly 454 includes the piston 460 and an infrared reflector 462, which defines one part of a position detector 464. According to the illustrated embodiment, the position detector 464 also includes a plurality of transmitter/sensor pairs 466, 468. According to the illustrated embodiment, the transmitters (or emitters) 466 may be in the form of infrared light emitting diodes, and the sensors 468 may be in the form of infrared sensors. According to other embodiments, the transmitters and sensors may use visible or ultraviolet light, for example. The transmitter/sensor pairs

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466, 468 are disposed along the length of the barrel 452 between the first end 456 and the second end 458. The reflector 462 may be in the form of a reflective strip that is disposed about the perimeter of a rigid disc 470 that is attached opposite the piston 460.

In operation, the position detector 464 (which could be coupled to the controller 150, for example) would use the interaction between the transmitter/sensor pairs 466, 468 and the reflector 462 to determine the position of the piston head assembly 454 along the barrel 452. In particular, light emitted from the transmitter 466 would be received by the sensor 468 (or would be received over a threshold amount) if the light contacts the reflector 462. Otherwise, the light would not be received by the sensor 468 (or would not be received below the threshold amount). Depending on the amount of light received by the sensor 468, a signal generated by the sensor 468 would vary. See, e.g., FIG. 13, wherein the signal of the sensor 468 varies as the piston head assembly 454 is drawn to the second end 458, with the sensor signal representative of first air, then piston O-ring, reflector 462 (corresponding to the peak in sensor output), piston, and finally fluid opposite the sensor 468. Depending on the signals received from the individual transmitter/sensor pairs 466, 468, the controller 150 may determine the position of the piston head assembly 454 along the barrel 452 between the first and second ends 456, 458.

A vacuum/pressure source (e.g., a diaphragm pump) 472 is attached via line (e.g., tubing) 474 to the end 458 of the barrel 452. The end 458 is otherwise closed, forming a first variable volume space 476 between the closed end 458 of the barrel 452 and the piston head assembly 454. Filtered air may be pumped into and out of the space 476 to cause the piston head assembly 454 to move between the first and second ends 456, 458 of the barrel 452. The movement of the piston head assembly 454 causes a second variable volume space 478 to open between the piston head assembly 454 and the first end 456 to receive fluid (e.g., a cell product) into the barrel 452. Compare FIGS. 11 and 12. Fluid may be drawn into (or may enter into) and pushed or delivered from the space 478 according to the movement of the piston head assembly 454.

In operation, the piston head assembly 454 starts at a first position, such as is illustrated in FIG. 11. The controller 150 causes the vacuum/pressure source to operate, and draw vacuum behind the piston head assembly 454 (i.e., in space 476). As a consequence, the piston head assembly 454 moves in the direction of the end 458 (i.e., from the end 456 to the end 458) and draws fluid into the space 478 (see FIG. 12). The controller 150 may subsequently operate the vacuum/pressure source to pump pressurized air into the space 476. This causes the piston head assembly 454 to move in the direction of the end 456 (i.e., from the end 458 to the end 456) and push fluid from the space 478.

It will be recognized that the pneumatic control of filtered air in and out of the space 476 provides certain advantages over the use of a syringe with a plunger arm where one end of the barrel remains open to the surrounding environment. By leaving the barrel end open, materials could collect on an inner surface of the barrel wall, such that movement of the piston head between the ends could permit the materials on the inner surface to interact with the fluid on the other (i.e., wet-side) of the piston head. The use of filtered air in the space 476 to move the piston 460 reduces or eliminates this potential source of contaminants. Further, the position detector 464 permits very precise control of the operation of the

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syringe pump 450. Other embodiments may use a mechanical or electro-mechanical mechanism to move the piston head 460, however.

FIGS. 14-16 illustrate an alternative embodiment of a syringe pump that may be used with either the first or the second syringe 116, 118 and as either the first and/or second syringe pump 142, 144.

The syringe pump is configured to use a syringe 490 with a wall defining a syringe barrel 492 (which may be made of cyclic olefin copolymer, or other materials such as may be inert, optically clear). Fluid may be drawn into a first end 494 of the syringe 490/syringe barrel 492, and may fill the barrel 492 to a second end 496 of the barrel 492. According to the illustrated embodiment of the syringe 490, the syringe lacks a piston or plunger head assembly moveable (translatable) between the first and second ends 494, 496; that is, the bore of the barrel 492 is without a piston or plunger disposed therein, or is plungerless. See FIGS. 15 and 16. According to other embodiments, an optional piston or plunger head assembly may be included.

The syringe 490 includes one or more capacitive sensors 498 that are configured to determine a location of an air/fluid interface (or, alternatively, the piston or plunger head assembly) within the barrel 492 of the syringe 490. As illustrated, the sensor 498 is a single sensor. The sensor 498 is illustrated in solid line in FIG. 14, and in dashed line in FIGS. 15 and 16. The sensor 498 may be in the form of a flexible, or flex, circuit that is disposed on an outer surface 500 of the barrel 492; for example, the sensor 498 may be attached to the outer surface of the barrel 492 using an adhesive.

So that the position of the sensor 498 relative to volumes defined within the barrel 492 may be more easily determined in FIGS. 15 and 16, the sensor 498 is not illustrated as extending from the first end 494 to the second end 496 of the barrel 492. In practice, it would be preferred that the sensor 498 extend entirely from the first end 494 of the barrel 492 to the second end 496 of the barrel 492. According to certain embodiments, however, there may be some amount of spacing permitted at one or both of the ends 494, 496 of the barrel 492.

The sensor 498 includes a pair of elements 502, 504 with a gap 506 disposed therebetween. The sensor 498 monitors the electrical capacity and senses distortions within the electrostatic field when other materials (other than air) are proximate to the sensor 498. As a consequence, the sensor 498 may determine the location of an air/fluid interface within the barrel 492 and/or the amount of fluid in the barrel 492. To better facilitate this determination, it is preferred that the barrel 492 be disposed with the barrel 492 oriented substantially vertically with respect to gravity and with the first end 494 oriented downwardly. See, e.g., FIG. 1.

A vacuum/pressure source (e.g., a diaphragm pump) 508 is attached via line (e.g., tubing) 510 to the end 496 of the barrel 492. The source 508 is used to draw fluid into the syringe 490, and to push the fluid from the syringe 490. As illustrated in FIGS. 15 and 16, fluid has been drawn into the syringe 490 such that an air/fluid interface 512 exists between a first volume or space 514 of the syringe 490 that is filled with fluid, and a second volume or space 516 of the syringe 490 that is filled with air. Because the end 496 is otherwise closed (by a cap with a filter, for example), the volume of both spaces 514, 516 may be varied by pumping air into and drawing air from the syringe 490, and in particular the space 516. The pumping of filtered air into and out of the space 516 causes the interface 512 to move between the first and second ends 494, 496 of the barrel 492.

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In operation, the interface **512** starts at a first position, such as may be slightly to the right of the position illustrated in FIG. **15**. The controller **150** causes the vacuum/pressure source **508** to operate, and draw vacuum from the syringe **490** (i.e., from space **516**). As a consequence, the interface **512** moves in the direction of the end **496** (i.e., from the end **494** to the end **496**) as fluid is drawn into the space **514** (i.e., from the state illustrated in FIG. **15** to that illustrated in FIG. **16**). The controller **150** may subsequently operate the vacuum/pressure source to pump pressurized air into the space **516**. This causes the interface **512** to move in the direction of the end **494** (i.e., from the end **496** to the end **494**) as fluid is pushed from the space **514** (i.e., from the state illustrated in FIG. **16** to that illustrated in FIG. **15**).

It will be recognized that the pneumatic control of filtered air in and out of the space **516** provides certain advantages over the use of a syringe with a plunger arm and associated piston or plunger head where one end of the barrel remains open to the surrounding environment, as mentioned above. By leaving the barrel end open, materials could collect on an inner surface of the barrel wall, such that movement of the piston between the ends could permit the materials on the inner surface to interact with the fluid on the other (i.e., wet-side) of the piston head. The use of filtered air in the space **516** to move an interface **512** similarly reduces or eliminates this potential source of contaminants. Moreover, in those embodiments where the piston or plunger head is omitted, any contamination attributable to the piston or plunger head itself may be prevented.

The sensor **498** may be coupled, for example, to the controller **150** to provide the position/volume information to the system **100**. In this regard, the sensor **498** may be coupled to a pair of connector pads disposed on the outer surface **500** of the syringe **490**. These connector pads may be disposed at the second end **496** of the syringe **490**; alternatively, the connector pads could be disposed elsewhere on the outer surface **500** of the syringe **490**. A pair of spring-biased or spring-loaded connectors (e.g., pogo pins) may be mounted on the hardware **104**, and may be configured to cooperate with the connector pads to couple the sensor **498** to the controller **150**.

As was mentioned above relative to the air/fluid interface sensors **380**, **382**, **384**, **386**, **388**, a syringe **490** (which may be pistonless or plungerless) incorporating a capacitive sensor **498** may be used not only as part of the syringe pump of the system **100** described herein, but the syringe **490** with sensor **498** may be used in other medical devices or systems that include a syringe pump, such as fill and finish systems illustrated and described in U.S. Pub. No. 2018/0155070 or the small volume cell processing systems described in U.S. Pub. No. 2017/0204371 or U.S. Pat. App. Pub. No. 2019/0351113, all of which are incorporated herein by reference.

Having discussed the structure of the illustrated embodiments of the fluid circuit **102** and the corresponding equipment of the reusable hardware **104**, the operation of the system **100** is now discussed with reference to FIG. **17**. As much of the operation of the system **100** involves control of the fluid flow between the containers **110**, **114**, **120**, **122**, **124**, **126**, **128**, the syringes **116**, **118**, and the spinning membrane **112** through the cassette **130**, reference is also made to FIGS. **1**, **2** and **6**.

A method **530** of operating the system **100** may begin with one or more pre-processing actions at blocks **532**, **534**, **536**. While each block **532**, **534**, **536** describes a different general type of activity, the actions at blocks **532**, **534**, **536** may include a plurality of individual actions. For example, priming the circuit **102** at block **536** may include individual

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actions of priming different portions of the circuit **102**, but for ease of illustration, the actions have been grouped together at block **536**.

Starting then with block **532**, the circuit **102** is installed on the hardware **104**. The controller **150** may determine when this is complete by checking one or more sensors, or the controller **150** may wait for an input to be received from the user via an input device, such as a touch screen. Once the controller **150** has determined that the circuit **102** is installed, the method passes to block **534**.

At block **534**, the controller **150** may perform numerous tests on the circuit **102** before any fluid is added to the circuit **102**. Certain tests, or checks, are performed on the syringe pumps **142**, **144**, other checks are performed on the clamps **320-329**. Because no fluid has been introduced to the circuit **102**, these checks may be referred to as dry checks. After the checks have been performed, the containers **122**, **124**, **126**, **128** containing the solutions for the process to be performed using the system may be connected to the circuit **102**, and in particular to the lines **166**, **168**, **176**, **180**.

The first set of actions may be performed on the syringes **116**, **118** and the syringe pumps **142**, **144**, and in particular those embodiments of the syringe pumps **142**, **144** including syringes with piston or plunger head assemblies and an optical sensor arrangement. As such, reference is made to the structures of the embodiment of FIGS. **11** and **12**. Initially, the pistons (e.g., **460**) of the syringes **116**, **118** are drawn to the second end of the syringe (e.g., end **458**, which end may be referred to as the upper end with the syringes **116**, **118** in the orientation of FIG. **1**) and then to the first end of the syringe (e.g., end **456**, which end may be referred to as the lower end with the syringes **116**, **118** in the orientation of FIG. **1**), which is done to permit sensor normalization to be conducted (typically either when the piston is at the second end or the first end). During these actions, all of the clamps **320-329** are left open. A circuit (or kit) integrity check is then performed by moving the pistons of the syringes **116**, **118** to the second end of the syringe with only clamps **320**, **321** closed. If the piston of either syringe **116**, **118** is able to move toward the second end, then this suggests a leak in the circuit **102** or that the clamp **321** is unable to maintain a vacuum.

After the first set of actions, further checks are performed on the clamps **320-329**. For example, clamps **322** and **325** may be closed, while the remaining clamps are open, and the piston of syringe **116** is moved toward the second end. Clamp **320** is then closed after a period of time, which should cause the piston in syringe **116** to stop moving and a negative pressure to build in the syringe **116**. A similar process can be conducted for other clamps. For example, clamps **321**, **322**, **326** can be tested with all clamps except clamps **321**, **322**, and **326** closed, and then each of clamp **321**, **322**, **326** closed some time after the piston of syringe **116** is moved toward the second end. For clamp **329**, all clamps **320-328** are closed and the piston of syringe **118** is moved toward the second end. The clamp **320** may also be checked again using the second syringe pump **144** and the second syringe **118**, with the process being generally the same except that clamps **320**, **322**, **325**, **326** are left open, and the piston of syringe **118** is advanced towards its second end.

Once the checks have been performed, the method **530** continues with the containers **110**, **122**, **124** and potentially container **126** being attached at the end of the actions of block **534** or the beginning of block **536**. With the containers **110**, **122**, **124** attached, the circuit **102** may be primed at block **536**.

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The priming of the circuit 102 may start with the priming of the fluid path to the second syringe pump 144. To do this, the controller 150 may open clamps 324, 326 (remainder closed) and cause the piston of the second syringe 118, in those embodiments that include a piston, to move toward the second end. In this and the remaining actions described below, where a piston is not present, this may be read instead as a reference to the movement of the air/fluid interface instead. This action draws wash fluid from the container 122, 124 through the port 174 and the channels 248, 250, 252 into the inlet 430 of the spinning membrane 112. The fluid passes through the spinning membrane 112, through port 210 and channels 236, 254 into the port 196 and syringe 118.

The fluid in the syringe 118 may be used to prime the path between the syringe pump 144 and the vent port 212 by closing all clamps except 321. The piston of the syringe 118 is then moved toward the first end to force fluid from the syringe 118 into channels 236, 234.

The priming of the circuit 102 may continue with the priming of the fluid path to the first syringe pump 142. To do this, the controller 150 opens clamps 322, 324 (remainder closed) and causes the piston of the first syringe 116 to move toward the second end. This draws wash fluid from the container 122, 124 through the port 174 and the channels 248, 250, 252 into the inlet 430 of the spinning membrane 112. The fluid passes through the spinning membrane 112, through port 208 and channels 232, 240, 242 and into the port 192 and syringe 116.

The fluid drawn into the first syringe 116 may be used to prime the fluid path to the source container 110. The controller 150 opens clamps 325, 327 and causes the piston of the first syringe 116 to move toward the first end. This pushes wash fluid from the syringe 116 through the port 192 and through channels 232, 244, 250, 256 to port 182. The fluid is able to pass from channel 232 to channel 244 because of the multiple apertures in clamp 322 that remain open to the chamber associated with that clamp even when the clamp 322 is closed. The fluid is pushed from the port 182 along the line 180 and into source container 110. This priming step removes air from the line 180 such that the system 100 is ready to begin processing cells.

The method 530 continues at block 538 with the controller 150 operating the system 100 to perform a procedure according to a protocol on the fluid in the source container 110. As one example, the controller 150 may operate the system 100 to separate cells from the fluid in the container 110, rinse the container 110 and wash the cells, and then pass the washed cells to container 128 for additional processing. As was the case with the actions at blocks 252, 254, 256, the actions at block 258 may include numerous individual actions, at least some of which may be repeated according to the amount of fluid in the source container 110, for example.

In this regard, a further flowchart is provided in FIG. 18 to illustrate the actions of the block 258 in FIG. 17. In general terms, the separation of the cells from the fluid in the container 150 requires that the cells be transferred from the container 110 to the spinning membrane 112, the spinning membrane 112 be operated to separate the cells from the filtrate, the filtrate transferred first to the syringe 116 and then to the container 120, and the cells transferred first to the syringe 118 and to the container 128, at least according to the illustrated embodiment. Certain of the actions illustrated in FIG. 18 follow in a necessary order, for example, operating the syringe pump 142 occurs as a consequence of a determination that the syringe 116 is full. On the other hand, certain actions may be performed in any order; for example,

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the determination if the syringe 116 is full may follow the determination if the spinning membrane 112 is full, rather than the order illustrated. Some of the actions may be optional, and an attempt has been made to represent optional actions with the use of dashed line.

To perform the separation of the cells from the fluid in the container 110 at block 550, clamps 322, 327 may be opened (remainder closed) and the piston of syringe 116 is moved toward the second end. This draws fluid from the source container 110 into the port 182, channels 250, 252, 256 and into inlet 430 of the spinning membrane 112. Fluid is drawn from the spinning membrane 112 through ports 208 and channels 232, 240, 242 into the port 192 and the syringe 116. While the fluid and cells are flowing into the spinning membrane 112, a rotor 436 of the spinning membrane 112 rotates at a separation rate defined by the protocol, and fluid is drawn from the spinning membrane 112 while the target cells accumulate in an annulus 438 of the spinning membrane 112 between an outer housing 440 of the spinning membrane 112 and an outer surface 442 of a membrane 444. See FIG. 10. The cells accumulate in the annulus 438 because the clamp 326 is closed.

Depending on the amount of fluid present in the container 110, it may be necessary to empty the syringe 116 from time to time into the filtrate container 120. This may be done when the syringe 110 is full or reaches a certain threshold volume, as determined at block 552. This may also be done when the system needs to reset the position the filtrate syringe piston at the first end (e.g., end 456), for example. At block 554, the controller 150 opens only clamp 320 while causing the piston of the syringe 116 to move in the direction of the first end, causing fluid to flow along the fluid path defined by channels 230, 232, port 164, and line 162 into the container 120.

Depending on the accumulation of the cells in the annulus 438 of the spinning membrane 112, it may be desirable to wash the cells in the spinning membrane 112 and move the cells to the syringe 118 (which may be referred to as harvesting the cells). See blocks 556, 558, 560. If the capacity of the spinning membrane 112 is not reached before the source container 110 is emptied (as determined at block 562, for example), then the cell wash may be conducted after other actions have occurred, namely the rinsing of the source container 110. See blocks 564, 566, 568, 570, 572. It is possible for the cell wash and harvest to be performed multiple times (e.g., once when the capacity of the spinning membrane 112 is reached at blocks 560, 562 and once after the source container 110 is emptied at block 570, 572). According to certain embodiments, the cell wash at blocks 558, 570 may not be performed, and the source rinse at blocks 566, 568 may also be optional.

To perform a cell wash (block 558, 570), the controller 150 opens the clamps 322, 324 (remainder closed) and controls the piston of the syringe 116 to draw fluid from the containers 122, 124 through the port 174 and channels 248, 250, 252, through the spinning membrane 112, port 208, and channels 232, 240, 242 and into the syringe 116. This causes the fluid originally surrounding the cells (which may be referred to as original supernatant) to be replaced with new solution (i.e., the wash media). To harvest the cells (block 560, 572), the controller 150 leaves clamp 324 open, closes clamp 322, and opens clamp 326. The piston of syringe 118 is caused to move toward the second end to draw fluid from the containers 122, 124 through the port 174 and channels 248, 250, 252, through the spinning membrane 112, port 210, and channels 236, 254 and into the syringe 118.

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When the source container 110 is emptied (block 562), the controller 150 may operate the system 100 to rinse the source container 110 (block 566), after which a cell wash and harvest is performed. To perform the rinse (block 568), the controller 150 first empties the syringe 116. Then, the controller 150 closes the clamp 320, and opens clamps 324, 325 (remainder closed) and moves the piston of the syringe 116 to draw fluid from the containers 122, 124 through port 174 and channels 232, 244, 248, 250 and into port 192 and syringe 116. Once a volume of the wash media has been drawn (or loaded) into the syringe 116, the syringe 116 is ready to deliver the rinse to the source container 110. To do so, the controller 150 leaves clamp 325 open, closes clamp 324, and opens clamp 327. The controller 150 then causes the piston of syringe 116 to push fluid through channels 232, 244, 250, 256 and port 182 into the source container 110. With the wash media transferred to the source container 110, the controller 150 can conduct a wash and harvest (blocks 570, 572).

At this point (block 574) or earlier if the syringe 118 is determined to be full (see blocks 576, 578), the contents of the syringe 118 may be transferred to the product container(s) 114. See also, block 540 of FIG. 17. Alternatively, according to the illustrated embodiment, the contents of the syringe 118 may be transferred to the container 128 at blocks 574, 578, where the concentration of the cells may be modified and other components may be added at blocks 542, 546, after which the product containers are filled at block 540. According to the illustrated embodiment, the component added may be a cryopreservation agent (or CPA).

To begin, the contents of the syringe 118 are transferred to the container 128. The controller 150 opens only clamp 329, and causes the piston of the syringe 118 to move in the direction of the first end. This pushes the contents of the syringe 118 through channels 236, 260 and port 186 into line 184 and container 128. The controller 150 then closes the clamp 329 and opens clamps 324, 326, and draws fluid into the syringe 118 through port 174 from containers 122, 124, into channels 248, 250, 252, the spinning membrane 112, and channels 236, 254, and to the port 196. The controller then closes clamps 324, 326, opens clamp 329, and causes the piston of syringe 118 to move in the opposite direction to push the contents (wash media) from the syringe 118 into container 128.

According to the illustrated embodiment, the controller 150 may pause the method 530 at block 544. In fact, the controller 150 may pause the method 530 twice: once to permit a sample to be drawn from the container 128, and a second time to permit the container 126 to be connected to the circuit 102 if the container 126 was not attached previously. Once the desired activities have been conducted, the method continues with the addition of the CPA.

As part of the addition of the CPA to the container 128, the controller 150 may first open only clamp 323 and cause the syringe 118 to draw CPA solution from the container 126 via line 176 and port 178 and through channels 236, 238, 246 into the syringe 118. At this point, the clamp 323 is closed, and clamp 321 may be opened to permit the syringe 118 to push any excess air from the syringe 118 and out the vent port 212 and filter 214 via channels 234, 236. Preferably, enough air is left in the syringe 118 to have an air chase from the second syringe pump 144 to the container 128. The controller 150 then closes clamp 321, opens clamp 323 and draws the desired volume of CPA solution from container 128 via line 176 and port 178 and through channels 236, 238, 246 into the syringe 118. The controller 150 closes

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clamp 323, opens clamp 329 and moves the piston of syringe 118 to push the CPA solution from the syringe 118 into the container 126.

As illustrated, the embodiment of the system 100 includes a table 600 (see FIGS. 1 and 3) on which the container 128 is disposed to oscillate therewith. The table 600 may be mounted on a motor-drive shaft 602 that permits the table 600 to oscillate about an axis 604. The controller 150 may control the table 600 (via the motor) to cause the table 600 to oscillate to agitate the contents of the container 128, encouraging mixing of the contents. This agitation may be performed, for example, while the CPA solution is being added to the container 128. The agitation may be continued for an additional time after the CPA solution has been added to encourage adequate mixing. The table 600 may also include a cooling or heating element that permits the material in the container 128 to be maintained at a particular temperature.

According to certain embodiments, the container 128 may be detached from the circuit 102. However, according to the illustrated embodiment, the contents of the container 128 are instead filled into one or more product containers 114 that are attached to the circuit 102. The system 100 may include a scale 606 (see FIGS. 1 and 3) for weighing the contents of the container(s) 114, although the sensitive nature of the volume control on the syringe pump 144 makes the use of the scale 606 more in the nature of a secondary check.

To begin the fill activity, the controller 160 opens the clamp 329 and causes the syringe pump 144 to draw fluid from the container 128 via line 184 and port 186 into channels 236, 260 to prime the fluid path between the container 126 and the pump 144. The controller 150 then closes the clamp 329 and opens clamp 321 to vent excess air from the syringe 118 via channels 234, 236. Preferably, an air chase volume remains in the syringe 118 if the final dose volume and air chase volume can be delivered in one syringe stroke. The air chase volume should be sufficient to fully move the final dose volume from the syringe 118 to the final container 114.

The controller 150 then closes clamp 321 and opens clamp 329 and causes the syringe pump 144 to draw fluid from the container 128 via line 184 and port 186 into channels 236, 260 to fill the syringe 118 with the desired volume of product. The controller 150 then closes clamp 329, opens clamp 328, and causes the syringe pump 144 to push fluid from the syringe 118 to one of the containers 114. The controller 150 may close the clamp 328 and open the clamp 321 to permit air to be drawn into the syringe 118, which air is then pushed from the syringe 118 with the clamp 321 closed and the clamp 328 open to provide an air chase volume to force the product solution into the container 114.

The system 100 may include other equipment as part of the hardware 104, in addition to the equipment already discussed, as illustrated in FIGS. 1 and 3. For example, the system 100 may include a display 608 with touch screen 610 to permit information to be entered into the system, including information regard the protocol of the procedure to be carried out by the system 100. The display 608 may be an electronic display, for example, with the touch screen 610 mounted thereon. Other input devices may be included, such as a pointer (e.g., mouse) and keyboard or keypad. As illustrated in FIGS. 1 and 3, an input device in the form of a barcode reader 612 may be attached to the system 100 to permit information to be inputted into the system 100 (and the controller 150) by scanning or reading a barcode, such as may be applied to the fluid circuit 102 or one or more of the containers 110, 114, 120, 122, 124, 126, 128. Other

output devices also may be included, such as one or more lights (e.g., light emitting diodes or bulbs) 614, which may be used to signal alerts, events or machine states to the operator.

The system 100 may also be used with one or more ancillary or secondary devices or peripherals, which peripherals may include valves, pumps, etc. to be used to control the filling of the product containers, for example. The system 100 (and in particular, the controller 150) may be in communication with the peripheral(s), and data may be transmitted back and forth between the system 100 and the peripheral(s) or may be shared between the system 100 and the peripheral(s). In fact, the peripheral(s) may have its own controller (as illustrated, which controller may include a microprocessor, other circuits or circuitry and one or more memories, which may be one or more tangible non-transitory computer readable memories, with computer executable instructions by which the microprocessor is programmed and which when executed by the microprocessor may cause the microprocessor to carry out one or more actions being stored on the memory/memories) that is in communication with the controller 150.

According to at least one embodiment, the ancillary or secondary device may be in the form of an external array of valves that can control the passage of fluid between the fluid circuit and the product containers, and the state of the valves may be controlled or triggered by the controller 150 via communication between the controller 150 and the controller associated with the array of valves.

Thus, an improved method and system for processing (e.g., concentrating or washing) small volumes of biological cells has been disclosed, in conjunction with an improved method and system for filling low-volume containers with the small volumes of processed. The description provided above, and the other aspects provided below, are intended for illustrative purposes, and are not intended to limit the scope of the disclosure to any particular method, system, apparatus or device described herein.

OTHER ASPECTS

Aspect 1. A fluid processing system comprising:

a disposable fluid circuit comprising:

a separation chamber; and

a flow control cassette comprising a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one interface sensor chamber in fluid communication with at least one of the plurality of separate channels, the at least one interface sensor chamber defined at least in part by a wall, and at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber; and

reusable hardware configured to accept the disposable fluid circuit and comprising:

a separator connected to the separation chamber;

a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor; and

at least one controller coupled to the separator, the at least one actuator and the capacitive sensor via the coupling, the controller configured to selectively operate the separator and the at least one actuator to provide a procedure according to a protocol.

Aspect 2. The fluid processing system according to aspect 1, wherein the at least one capacitive sensor comprises a flexible circuit.

Aspect 3. The fluid processing system according to aspect 1 or 2, wherein the at least one capacitive sensor comprises a connector pad, and the coupling comprises a spring-biased connector.

Aspect 4. The fluid processing system according to aspect 3, wherein the spring-biased connector comprises a pogo pin.

Aspect 5. The fluid processing system according to any one of aspects 1-4, wherein the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the separator comprises a spinning membrane separator drive coupled to the spinning membrane separator.

Aspect 6. The fluid processing system according to any one of aspects 1-5, wherein:

the disposable fluid circuit comprises first and second syringes, and

the reusable hardware comprises first and second syringe pumps, the first and second syringes coupled to the first and second syringe pumps respectively, the first syringe pump configured to move fluid into and out of the first syringe and the second syringe pump configured to move fluid into and out of the second syringe.

Aspect 7. The fluid processing system according to any one of aspects 1-6, wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining a plurality of fluid paths.

Aspect 8. The fluid processing system according to any one of aspects 1-4, wherein:

the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the separator comprises a spinning membrane separator drive coupled to the spinning membrane separator;

the disposable fluid circuit comprises first and second syringes, and the reusable hardware comprises first and second syringe pumps, the first and second syringes coupled to the first and second syringe pumps respectively, the first syringe pump configured to move fluid into and out of the first syringe and the second syringe pump configured to move fluid into and out of the second syringe;

the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining at least a first path between a source container, the spinning membrane, the filtrate outlet and the first syringe, a second path between the first syringe and a filtrate container, and a third path between a wash media container, the spinning membrane inlet, the retentate outlet and the second syringe; and

the at least one controller coupled to the spinning membrane separator drive, the first and second syringe pumps, the at least one actuator, and the capacitive sensor via the coupling, the controller configured to selectively operate the spinning membrane separator drive, the first and second syringe pumps, and the at least one actuator to provide a procedure according to a protocol.

Aspect 9. The fluid processing system according to aspect 8, wherein:

the first syringe pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered

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air into and out of the first syringe to draw fluid into and push fluid from the first syringe; and
the second syringe pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered air into and out of the second syringe to draw fluid into and push fluid from the second syringe.

Aspect 10. A fluid processing system comprising:

a disposable fluid circuit comprising:

a separation chamber;

at least one plungerless syringe, the syringe comprising a wall defining a barrel having a first end and a second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the syringe; and

a flow control cassette comprising a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one of the plurality of separate channels connected to the first end of the barrel of the at least one plungerless syringe; and

reusable hardware configured to accept the disposable fluid circuit and comprising:

a separator connected to the separation chamber;

at least one syringe pump, the second end of the barrel of the at least one syringe coupled to the at least one syringe pump, the at least one syringe pump configured to draw fluid into and push fluid from the at least one syringe;

a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor; and
at least one controller coupled to the separator, the at least one syringe pump and the control cassette interface, the controller configured to selectively operate the separator, the at least one syringe pump and the interface to provide a procedure according to a protocol.

Aspect 11. The fluid processing system according to aspect 10, wherein the at least one capacitive sensor comprises a single capacitive sensor disposed on the outer surface of the at least one syringe from the first end to the second end of the barrel.

Aspect 12. The fluid processing system according to aspect 10 or 11, wherein the capacitive sensor comprises a flexible circuit.

Aspect 13. The fluid processing system according to any one of aspects 10-12, wherein the capacitive sensor comprises a connector pad, and the coupling comprises a spring-biased connector.

Aspect 14. The fluid processing system according to aspect 13, wherein the spring-biased connector comprises a pogo pin.

Aspect 15. The fluid processing system according to any one of aspects 10-14, wherein the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the separator comprises a spinning membrane separator drive coupled to the spinning membrane separator.

Aspect 16. The fluid processing system according to any one of aspects 10-15, wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining a plurality of fluid paths.

Aspect 17. The fluid processing system according to any one of aspects 10-16, wherein:

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the at least one syringe comprises a first syringe and a second syringe, and

at least one syringe pump comprises a first syringe pump and a second syringe pump, the second end of the barrel of the first syringe coupled to the first syringe pump, the first syringe pump configured to draw fluid into and push fluid from the first syringe, and the second end of the barrel of the second syringe coupled to the second syringe pump, the second syringe pump configured to draw fluid into and push fluid from the second syringe.

Aspect 18. The fluid processing system according to aspect 17, wherein:

the first syringe pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered air into and out of the first syringe to draw fluid into and push fluid from the first syringe; and

the second syringe pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered air into and out of the second syringe to draw fluid into and push fluid from the second syringe.

Aspect 19. The fluid processing system according to aspect 18, wherein:

the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the reusable hardware comprises a spinning membrane separator drive coupled to the spinning membrane separator;

wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining at least a first path between a source container, the spinning membrane, the filtrate outlet and the first syringe, a second path between the first syringe and a filtrate container, and a third path between a wash media container, the spinning membrane inlet, the retentate outlet and the second syringe; and

at least one controller coupled to the spinning membrane separator drive, the first and second syringe pumps, the at least one actuator, and the capacitive sensor via the coupling, the controller configured to selectively operate the spinning membrane separator drive, the first and second syringe pumps, and the at least one actuator to provide a procedure according to a protocol.

Aspect 20. A fluid processing system comprising:

a disposable fluid circuit comprising:

a separation chamber;

at least one plungerless syringe, the syringe comprising a wall defining a barrel having a first end and a second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the syringe; and

a flow control cassette comprising a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one of the plurality of separate channels connected to the first end of the barrel of the at least one plungerless syringe, at least one interface sensor chamber in fluid communication with at least one of the plurality of separate channels, the at least one interface sensor chamber defined at least in part by a wall, and at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber; and

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reusable hardware configured to accept the disposable fluid circuit and comprising:

- a separator connected to the separation chamber;
- at least one syringe pump, the second end of the barrel of the at least one syringe coupled to the at least one syringe pump, the at least one syringe pump configured to draw fluid into and push fluid from the at least one syringe;
- a coupling connected to the at least one capacitive sensor disposed on an outer surface of the syringe;
- a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber;
- at least one controller coupled to the separator, the at least one syringe pump and the control cassette interface, the controller configured to selectively operate the separator, the at least one syringe pump and the interface to provide a procedure according to a protocol.

Aspect 21. The fluid processing system according to aspect 20, wherein the at least one capacitive sensor disposed on an outer surface of the syringe and the at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber each comprises a flexible circuit.

Aspect 22. The fluid processing system according to aspect 20 or 21, wherein the at least one capacitive sensor disposed on an outer surface of the syringe and the at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber each comprises a connector pad, and the couplings each comprises a spring-biased connector.

Aspect 23. The fluid processing system according to aspect 22, wherein the spring-biased connector comprises a pogo pin.

Aspect 24. The fluid processing system according to any one of aspects 20-23, wherein the at least one capacitive sensor disposed on an outer surface of the syringe comprises a single capacitive sensor disposed on the outer surface of the syringe from the first end to the second end of the barrel.

Aspect 25. The fluid processing system according to any one of aspects 20-24, wherein the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the separator comprises a spinning membrane separator drive coupled to the spinning membrane separator.

Aspect 26. The fluid processing system according to any one of aspects 20-25, wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining a plurality of fluid paths.

Aspect 27. The fluid processing system according to any one of aspects 20-26, wherein:

- the at least one syringe comprises a first syringe and a second syringe, and
- at least one syringe pump comprises a first syringe pump and a second syringe pump, the second end of the barrel of the first syringe coupled to the first syringe pump, the first syringe pump configured to draw fluid into and push fluid from the first syringe, and the second end of the barrel of the second syringe coupled to the second syringe pump, the second syringe pump configured to draw fluid into and push fluid from the second syringe.

Aspect 28. The fluid processing system according to aspect 27, wherein:

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the first syringe pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered air into and out of the first syringe to draw fluid into and push fluid from the first syringe; and

the second syringe pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered air into and out of the second syringe to draw fluid into and push fluid from the second syringe.

Aspect 29. The fluid processing system according to aspect 28, wherein:

the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the reusable hardware comprises a spinning membrane separator drive coupled to the spinning membrane separator;

wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining at least a first path between a source container, the spinning membrane, the filtrate outlet and the first syringe, a second path between the first syringe and a filtrate container, and a third path between a wash media container, the spinning membrane inlet, the retentate outlet and the second syringe; and

the at least one controller coupled to the spinning membrane separator drive, the first and second syringe pumps, the at least one actuator, and the capacitive sensor via the coupling, the controller configured to selectively operate the spinning membrane separator drive, the first and second syringe pumps, and the at least one actuator to provide a procedure according to a protocol.

The invention claimed is:

1. A fluid processing system comprising:

a disposable fluid circuit comprising:

a separation chamber;

at least one plungerless syringe, the at least one plungerless syringe oriented vertically and comprising a wall defining a barrel having a first end and a second end with the first end being located below the second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the at least one plungerless syringe;

a flow control cassette comprising a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one of the plurality of separate channels connected to the first end of the barrel of the at least one plungerless syringe;

the flow control cassette further comprising at least one separately defined interface sensor chamber in fluid communication with at least one of the plurality of separate channels connected to the first end of the barrel of the at least one plungerless syringe, the at least one separately defined interface sensor chamber further defined at least in part by a wall separate from the at least one of the plurality of separate channels, and at least one capacitive sensor disposed on the wall of the at least one separately defined interface sensor chamber to detect the presence of air; and

reusable hardware configured to accept the disposable fluid circuit and comprising:

a separator connected to the separation chamber of the disposable fluid circuit;

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at least one vacuum/pressure air source coupled to the second end of the barrel of the at least one plungerless syringe of the disposable fluid circuit, the at least one vacuum/pressure air source configured to pump filtered air into and out of the at least one plungerless syringe so as to form at least one syringe pump configured to draw fluid into and push fluid from the at least one plungerless syringe and selectively through the plurality of separate channels connected to the plurality of selectable junctions within the flow control cassette;

a control cassette interface being coupled to the at least one capacitive sensor disposed on the wall of the at least one separately defined interface sensor chamber when the flow control cassette of the disposable fluid circuit is installed on the reusable hardware, with the control cassette interface having at least one actuator for each of the selectable junctions of the flow control cassette, and being coupled to the at least one capacitive sensor disposed on the outer surface of the wall of the at least one plungerless syringe; and

at least one controller coupled to the separator, the at least one vacuum/pressure air source and the control cassette interface; the at least one controller configured to selectively operate the separator, the at least one vacuum/pressure air source and the control cassette interface to process a fluid according to a protocol.

2. The fluid processing system according to claim 1, wherein the at least one capacitive sensor on the outer surface of the wall of the at least one plungerless syringe comprises a single capacitive sensor disposed on the outer surface of the at least one plungerless syringe from the first end to the second end of the barrel.

3. The fluid processing system according to claim 1, wherein the capacitive sensor on the outer surface of the wall of the at least one plungerless syringe comprises a flexible circuit.

4. The fluid processing system according to claim 1, wherein the at least one capacitive sensor disposed on the wall of the at least one separately defined interface sensor chamber comprises a flexible circuit.

5. The fluid processing system according to claim 1, wherein the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the separator comprises a spinning membrane separator drive coupled to the spinning membrane separator.

6. The fluid processing system according to claim 1, wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable

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junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining a plurality of fluid paths.

7. The fluid processing system according to claim 1, with the disposable fluid circuit further comprising:

a second plungerless syringe, the second plungerless syringe oriented vertically and comprising a wall defining a barrel having a first end and a second end with the first end being located below the second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the second plungerless syringe, and with the reusable hardware further comprising:

a second vacuum/pressure air source coupled to the second end of the barrel of the second plungerless syringe of the disposable fluid circuit, the second vacuum/pressure air source configured to pump filtered air into and out of the second plungerless syringe so as to form a second syringe pump configured to draw fluid into and push fluid from the second plungerless syringe and selectively through the plurality of separate channels connected to the plurality of selectable junctions within the flow control cassette.

8. The fluid processing system according to claim 7, wherein:

the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the reusable hardware comprises a spinning membrane separator drive coupled to the spinning membrane separator;

wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining at least a first path between a source container, the spinning membrane separator, the filtrate outlet and the first plungerless syringe, a second path between the first plungerless syringe and a filtrate container, and a third path between a wash media container, the spinning membrane separator inlet, the retentate outlet and the second plungerless syringe; and the at least one controller coupled to the spinning membrane separator drive, the first and second syringe pumps, the at least one actuator, and the at least one capacitive sensor on the outer surface of the wall of the first plungerless syringe; the at least one controller configured to selectively operate the spinning membrane separator drive, the first and second syringe pumps, and the at least one actuator to provide a procedure according to a protocol.

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